Real-Time Web Communication Standards: A Retrospective and Projection 2021–2031

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Executive Summary

- **Late 2021:** Only WebRTC was a finalized standard with broad browser support, and even it had major cross-browser gaps (Safari: H.264 only, no simulcast, broken iOS DataChannel). QUIC was newly standardized but not universal. HTTP/3 was pre-RFC. WebTransport was non-functional. WebCodecs existed in one browser. MoQ did not exist.
- **March 2026:** The core stack (WebRTC, QUIC, HTTP/3) is standardized and universal. WebCodecs is cross-browser with 94% coverage. Two gaps remain: Safari lacks WebTransport (82% coverage) and Encoded Transforms for E2EE remain Chrome-only.
- **2028–2029 projection:** WebTransport reaches Safari via Interop 2026. Encoded Transforms become cross-browser. WebCodecs achieves W3C Recommendation. AV1 becomes a practical WebRTC codec as hardware encoding expands. MoQ and RTP over QUIC reach RFC status.
- **2029–2031 projection:** Full stack convergence. The "unbundled" architecture (WebTransport + WebCodecs + WebAssembly) becomes viable for specialized use cases. AV1 hardware encoding is ubiquitous. MoQ enables standards-based sub-second streaming with CDN fan-out.
- **Hardware and network:** Standards maturation compounds with hardware improvement (60–80% CPU gains vs. 2021, ubiquitous AV1 silicon) and network growth (200–300 Mbps fixed, 100+ Mbps mobile). By 2030, a 720p call consumes less than 0.5% of link capacity.
- **Information-theoretic view (Appendix ??):** Codec efficiency gaps narrow from H.264 ($\Delta \approx 2.1$) to AV1 ($\Delta \approx 0.4$), approaching the Shannon rate-distortion bound. QUIC eliminates head-of-line blocking, recovering up to 34% per-stream throughput at 5% loss. WebTransport's unreliable datagrams outperform WebSocket for any application with a freshness half-life below 34 ms.

1 Introduction

This document examines the concrete browser capabilities enabled by the real-time web communication standards collected in the `standards/` directory of this project. It compares what these standards made possible in browsers as of late 2021 with what they enable as of March 2026, then projects where the stack is heading through 2031. The discussion focuses exclusively on what standards adoption in browsers enabled and will enable — not on market trends, product adoption, or commercial developments.

The standards under review span five areas:

- **WebRTC** — W3C WebRTC 1.0 and supporting IETF protocols (RFCs 8829, 8445, 8826, 8827, 3550, 3711)
- **QUIC** — RFC 9000
- **HTTP/3** — RFC 9114
- **WebTransport** — W3C Working Draft and RFC 9297
- **WebCodecs** — W3C Working Draft

Together, these standards define a full stack for real-time communication in the browser: from low-level transport (QUIC) through application-layer protocols (HTTP/3, WebTransport) to peer-to-peer media and data exchange (WebRTC) and direct codec access (WebCodecs).

Sections 3–7 detail each standard's browser capabilities in late 2021 and March 2026. Sections 8–9 project the stack's evolution through 2031, grounded in active W3C charter milestones, IETF draft timelines, browser vendor signals, and hardware and network trends. Section 10 summarizes the full 2021–2031 trajectory. Appendix **??** applies information-theoretic measures to quantify channel capacity, source coding efficiency, and effective information rates across the same time horizons.

Section 2 first provides a concise overview of what each standard does and how the pieces fit together, for readers unfamiliar with the protocol landscape.

2 The Standards Stack

The standards under review form a layered architecture built on two complementary transport paths—one *peer-to-peer* (browser talks directly to browser) and one *client-server* (browser talks to a server)—with a shared codec layer that bridges both. This section explains what each standard does, what problem it solves, and how the pieces fit together. Figure 1 provides a visual map; the subsections below walk through each layer from the bottom up.

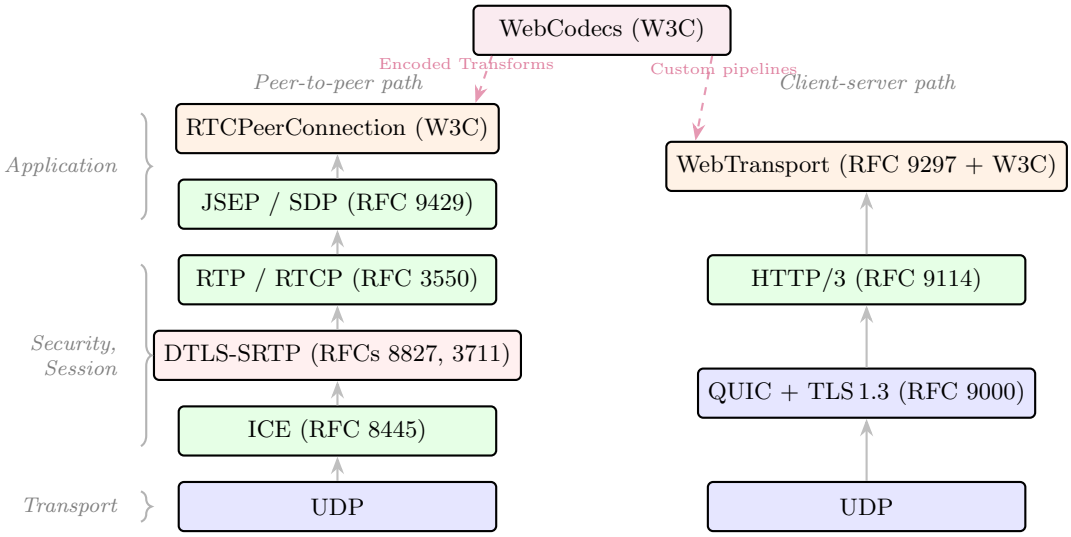


Figure 1: Protocol stack for the real-time web communication standards. Left: the WebRTC peer-to-peer path. Right: the QUIC-based client-server path. WebCodecs (top centre) provides frame-level codec access regardless of transport.

2.1 The Underlying Transport: UDP

All the real-time protocols discussed here ultimately send their data over **UDP (User Datagram Protocol)**. UDP is intentionally simple: it sends individual packets (called *datagrams*) from one machine to another with no guarantees about ordering, delivery, or retransmission. If a packet is lost, UDP does not notice and does not resend it.

This might seem like a poor foundation, but for real-time media it is actually an advantage. In a video call, a video frame that arrives 200 ms late is useless—the moment it was needed has already

passed. The alternative, **TCP (Transmission Control Protocol)**, guarantees that every byte arrives in order by automatically retransmitting lost packets, but this creates a problem called *head-of-line blocking*: while TCP waits for a lost packet to be retransmitted, it holds up delivery of all subsequent data, even data for completely independent streams. For interactive media, this stalling is worse than simply dropping the lost packet and moving on. UDP lets the protocols built on top of it make their own choices about what to retransmit and what to discard.

2.2 WebRTC: Peer-to-Peer Media in the Browser

WebRTC (Web Real-Time Communication) is a W3C standard that gives browsers a built-in ability to send audio, video, and arbitrary data directly from one browser to another—peer-to-peer, without routing media through a server. The browser exposes this through a single JavaScript API called `RTCPeerConnection`. Under the hood, `RTCPeerConnection` orchestrates a set of IETF protocols that handle four problems: *finding a network path* between two peers, *agreeing what media to send*, *encrypting the media*, and *actually carrying it*. The subsections below explain each problem and the protocol that solves it.

2.2.1 Finding a Path: ICE, STUN, and TURN

The first problem is deceptively hard. Most devices on the internet do not have their own public IP address. Instead, they sit behind a **NAT (Network Address Translator)**—typically the home router or corporate firewall—which assigns the device a *private* address (like `192.168.1.x`) and translates outgoing traffic onto a single *public* address that is visible to the rest of the internet. This means two browsers behind different NATs cannot simply send packets to each other: neither knows the other's routable address, and even if it did, the receiving NAT would likely drop the unsolicited incoming packet.

WebRTC solves this with **ICE (Interactive Connectivity Establishment, RFC 8445)**. ICE is not a server—it is an algorithm that runs in each browser, systematically searching for a working network path between two peers. The search uses two types of infrastructure servers:

STUN (Session Traversal Utilities for NAT) is a lightweight, stateless server that serves one purpose: when a browser sends it a request, the STUN server replies with the browser's *public* IP address and port—the address as seen from the far side of the NAT. This is called a *server-reflexive candidate*: the browser now knows what address the outside world sees for it. The STUN server relays no media and maintains no per-session state; it simply reflects the address back.

TURN (Traversal Using Relays around NAT) is a heavier-weight server used as a last resort. When two peers are both behind restrictive NATs (such as corporate firewalls that block all unsolicited inbound traffic), no direct path exists between them. In this case, both peers connect to a TURN server, which allocates ports, authenticates each peer, and *relays* all media packets between them for the duration of the session. This works through any NAT because both peers are making outbound connections to the TURN server. The cost is that all media flows through the relay, adding latency and bandwidth consumption. Approximately 10–20% of WebRTC connections require TURN.

The ICE process works as follows. Each browser gathers a set of *candidates*—potential addresses through which it might be reachable:

1. **Host candidates:** the device's own local IP addresses (useful when both peers are on the same local network).
2. **Server-reflexive candidates:** the public address discovered via STUN (useful when the NAT allows direct traffic once the mapping is known).
3. **Relay candidates:** a port allocated on a TURN server (the guaranteed fallback).

Both peers exchange their candidate lists (via the signaling mechanism described below), and ICE constructs every possible pairing of one local candidate with one remote candidate. It then tests each pair in priority order—host pairs first (lowest latency), then server-reflexive pairs, then relay pairs—by sending a probe and waiting for a response. The first pair that works is selected, and media begins flowing over it.

Trickle ICE (RFC 8838) improves this process by allowing candidates to be exchanged and tested incrementally, as they are discovered, rather than waiting for the full list to be compiled. In practice, this reduces connection setup time to roughly 500–800 ms because the fastest candidates (host and server-reflexive) are typically discovered and tested within the first few hundred milliseconds.

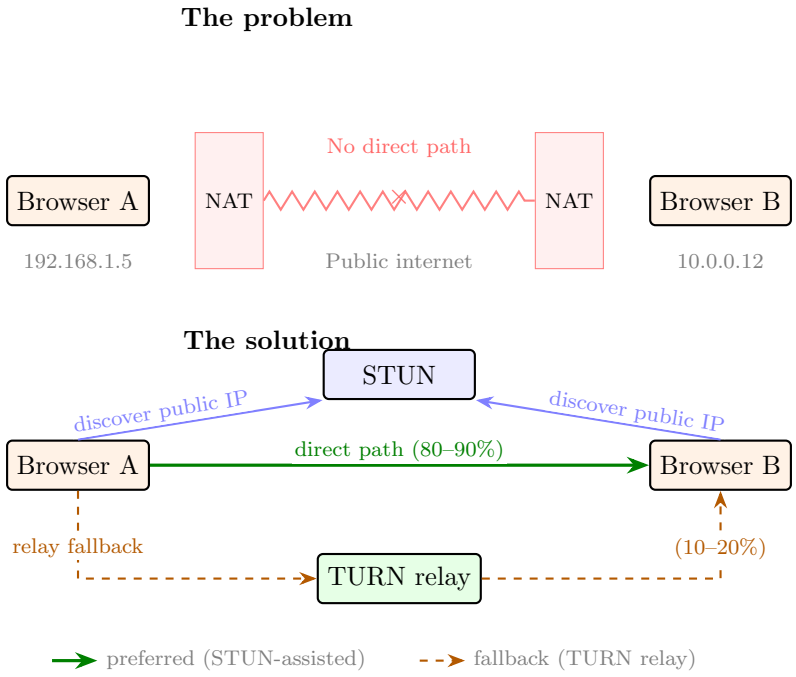


Figure 2: The NAT problem and how ICE solves it. Top: two browsers behind NATs cannot reach each other directly—neither knows the other's public address. Bottom: STUN discovers each browser's public address; ICE attempts a direct connection (successful 80–90% of the time) and falls back to a TURN relay when direct connectivity is impossible.

2.2.2 Agreeing What to Send: SDP, JSEP, and Signaling

Before media can flow, two peers must agree on what they will send each other: which codecs to use (H.264, VP9, AV1), which media types (audio, video, screen-share), at what resolutions and bitrates, and which encryption parameters. This negotiation uses the **Session Description Protocol (SDP)**—a text-based format that describes a peer's media capabilities.

The process works through an *offer/answer* exchange:

1. **Peer A creates an SDP offer** listing everything it can send and receive: "I support H.264 and VP9 video, Opus audio, and here is my encryption certificate fingerprint."
2. **The offer is delivered to Peer B** through a signaling channel.
3. **Peer B creates an SDP answer** selecting the common ground: "I also support VP9 and Opus; let's use those. Here is my encryption fingerprint."
4. **The answer is delivered back to Peer A.** Both peers now know exactly what media they will exchange and how it will be encrypted.

JSEP (JavaScript Session Establishment Protocol, RFC 9429; originally RFC 8829) defines how the browser creates and processes these SDP messages. It specifies the JavaScript API calls—`createOffer()`, `createAnswer()`, `setLocalDescription()`, `setRemoteDescription()`—that applications use to drive the negotiation.

Critically, JSEP *deliberately does not specify how* the SDP offer and answer are delivered between the two peers. This delivery mechanism is called **signaling**, and it is the one part of a WebRTC system that the application must build itself. The signaling channel can be anything: a WebSocket connection, an HTTP request, a message passed through a shared database—whatever gets the SDP text from one peer to the other. The signaling server also relays ICE candidates as they are discovered (for Trickle ICE). Importantly, the signaling server never sees or touches the actual media; it handles only lightweight control messages.

2.2.3 Encrypting the Media: DTLS and SRTP

All WebRTC media is encrypted. There is no option to send unencrypted audio or video—encryption is mandatory by design.

The encryption uses two protocols working together:

DTLS (Datagram Transport Layer Security, RFC 8827) is essentially TLS (the same encryption used for HTTPS websites) adapted for UDP. Once ICE has found a working path between two peers, DTLS performs a cryptographic handshake over that path. During the handshake, each peer presents a self-signed certificate, and the remote peer verifies that the certificate matches the fingerprint it received earlier in the SDP exchange. This binding between the signaling channel and the media channel prevents *man-in-the-middle attacks*: an attacker cannot substitute their own encryption keys without the mismatch being detected. The DTLS handshake produces a shared set of encryption keys.

SRTP (Secure Real-time Transport Protocol, RFC 3711) uses the keys negotiated by DTLS to encrypt every individual media packet. SRTP adds a small authentication tag to each packet (a few bytes), enabling the receiver to verify that the packet has not been tampered with in transit. The overhead is negligible and introduces no meaningful latency.

2.2.4 Carrying the Media: RTP and RTCP

With a path established (ICE), capabilities agreed (SDP/JSEP), and encryption keys negotiated (DTLS), media can finally flow. The protocol that carries the actual audio and video is **RTP (Real-time Transport Protocol, RFC 3550)**, sent as encrypted SRTP packets.

RTP provides the essential bookkeeping for real-time media:

- **Sequence numbers** so the receiver can detect lost packets and reorder packets that arrive out of sequence.
- **Timestamps** so the receiver can synchronize audio and video playback (lip sync) and play back frames at the correct rate.
- **Payload type identifiers** so the receiver knows which codec was used to encode each packet.

Alongside RTP, the **RTP Control Protocol (RTCP)** provides a feedback channel. Receivers periodically send RTCP reports back to the sender with statistics on packet loss rate, jitter (variation in packet arrival times), and round-trip time. The sender uses this feedback to adapt in real time: if packet loss increases, the video encoder can reduce its bitrate to avoid further congestion. This adaptive behaviour is essential for maintaining call quality over varying network conditions.

2.2.5 The WebRTC Security Model

RFC 8826 defines the overarching security principles for WebRTC:

- **Mandatory encryption:** as described above, all media is encrypted via DTLS-SRTP.
- **User consent:** the browser must obtain explicit user permission before accessing the camera or microphone. A web page cannot silently record audio or video.
- **IP address privacy:** browsers use **mDNS (multicast DNS)** to mask local IP addresses in ICE candidates. Without this, a web page could discover the user's local network address merely by initiating a WebRTC connection, creating a privacy leak. With mDNS, the browser substitutes a random hostname that resolves only on the local network.

2.2.6 Putting It Together: A WebRTC Call Sequence

Figure 3 illustrates the end-to-end sequence. When a user joins a video call:

1. The browser gathers ICE candidates (local addresses, STUN-discovered public addresses, and optionally TURN relay addresses).
2. The application creates an SDP offer describing the browser's media capabilities and sends it, along with the ICE candidates, through the signaling server to the remote peer.
3. The remote peer responds with an SDP answer and its own ICE candidates.
4. ICE tests candidate pairs and selects the best working path.
5. DTLS performs a handshake over that path to negotiate encryption keys.
6. Encrypted media (audio and video as SRTP packets) begins flowing directly between the two browsers. The signaling server is no longer in the media path.

2.3 QUIC: A Modern Transport Protocol

QUIC (RFC 9000) is a general-purpose transport protocol that, like WebRTC's media path, runs over UDP. It was originally developed by Google and standardized by the IETF in 2021. QUIC is designed to solve several long-standing problems with TCP, the transport protocol that has carried most internet traffic for decades.

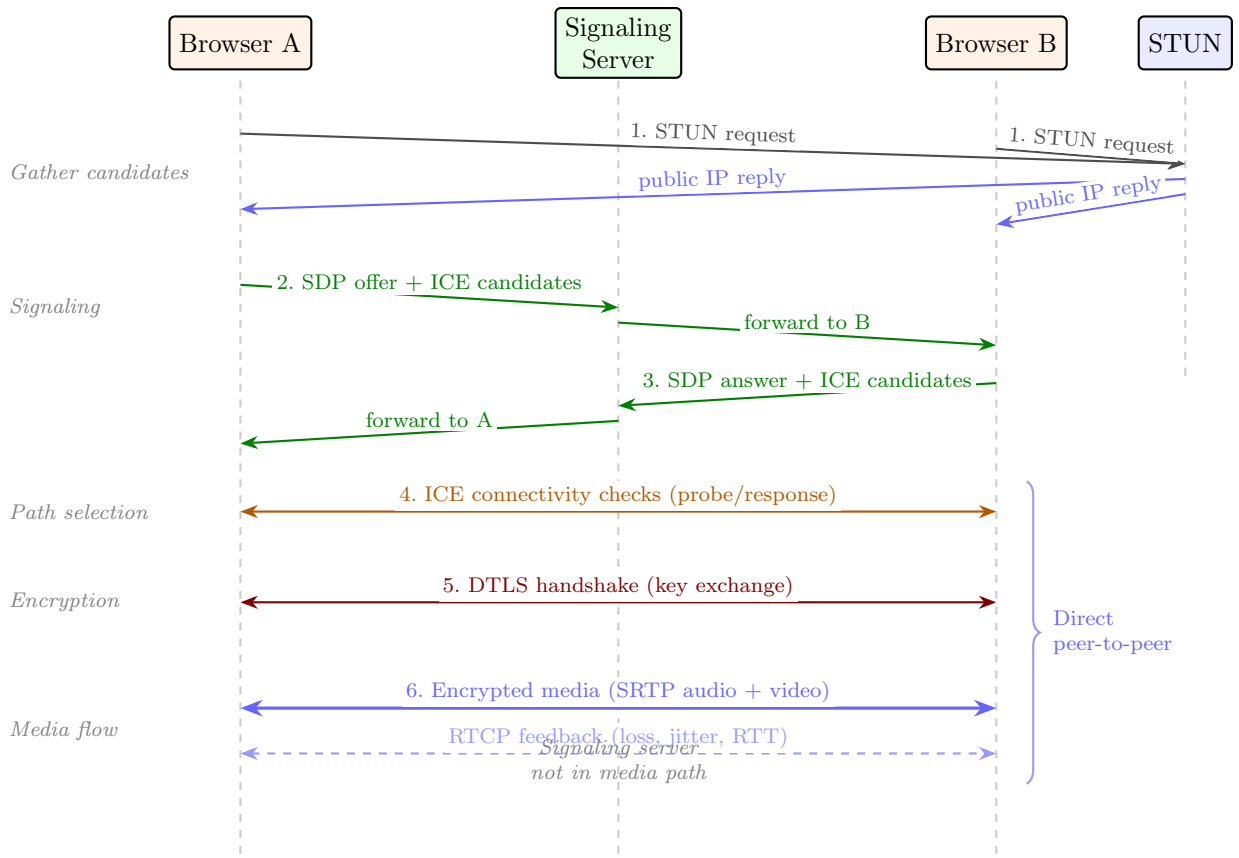


Figure 3: WebRTC call establishment sequence. Steps 1–3 use supporting infrastructure (STUN server, signaling server). Once ICE selects a path and DTLS establishes encryption (steps 4–5), media flows directly between the two browsers (step 6). The signaling server plays no further role in the media path.

The head-of-line blocking problem. TCP provides a single, ordered byte stream: every byte must be delivered in sequence. When a browser opens multiple parallel data flows over one TCP connection (as HTTP/2 does), a single lost packet blocks *all* of those flows until the lost packet is retransmitted and received. This is head-of-line blocking. QUIC solves it by providing *many independent streams* within a single connection, each with its own sequencing and flow control. A lost packet on one stream stalls only that stream; all others continue unaffected. For applications carrying multiple concurrent flows—such as a video call with several media tracks—this independence is critical.

Built-in encryption. QUIC integrates TLS 1.3 (Transport Layer Security, the same encryption standard used for HTTPS) directly into the connection handshake. A new QUIC connection is established in a single network round trip (1-RTT), compared to TCP's three-way handshake followed by a separate TLS handshake (typically 2–3 round trips). For servers the client has previously connected to, QUIC can establish a connection with *zero* round trips (0-RTT), sending application data in the very first packet.

Connection migration. Traditional TCP connections are identified by a *5-tuple*: source IP address, source port, destination IP address, destination port, and protocol number. When any of these change—for example, when a phone switches from Wi-Fi to cellular—the TCP connection

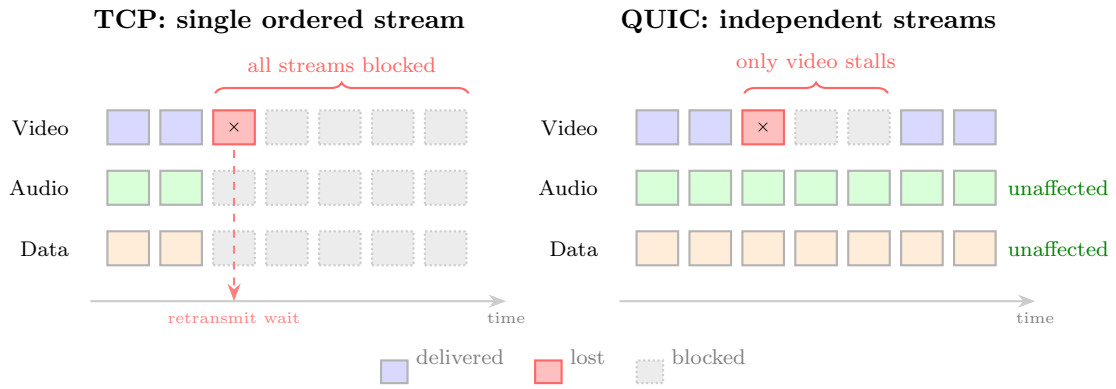


Figure 4: Head-of-line blocking: TCP vs. QUIC. Left: when a TCP packet is lost, *all* multiplexed streams stall until the lost packet is retransmitted. Right: QUIC isolates the loss to the affected stream; audio and data continue flowing without interruption.

breaks and must be re-established. QUIC instead identifies connections by *connection IDs*, opaque tokens that survive network changes. When a device changes networks, QUIC validates the new path and continues the session without interruption. For a video call, this means seamless handoff rather than a disconnect-and-reconnect.

2.4 HTTP/3: HTTP over QUIC

HTTP/3 (RFC 9114) is the latest version of the HTTP protocol that powers the web. It maps HTTP requests and responses onto QUIC instead of TCP. Each HTTP request/response pair occupies its own QUIC stream, so the head-of-line blocking that affects HTTP/2 (which multiplexes over a single TCP connection) is eliminated: a lost packet delays only the request it belongs to, not every other request on the connection.

For the standards discussed in this document, HTTP/3 matters primarily because it is the foundation on which WebTransport is built. A WebTransport session is established over an HTTP/3 connection, inheriting all of QUIC's properties: independent streams, built-in encryption, connection migration, and low-latency connection establishment.

2.5 WebTransport: QUIC Capabilities for Browser Applications

WebTransport (W3C Working Draft + RFC 9297) is a browser API that gives JavaScript applications direct access to QUIC's transport capabilities over an HTTP/3 connection. To understand why this matters, consider the alternative: **WebSocket**.

WebSocket, the current standard for bidirectional browser-server communication, runs over TCP. It provides a single ordered byte stream— the same model as a regular TCP connection, with the same head-of-line blocking problem. If a WebSocket sends a burst of small messages and one packet is lost, all subsequent messages stall until the lost one is retransmitted. Furthermore, every message sent over WebSocket is guaranteed to arrive (TCP will retransmit until it does), even if the message has become stale by the time it finally arrives.

WebTransport offers three capabilities that WebSocket cannot:

- **Unreliable datagrams.** An application can send a message that is delivered if possible but

silently dropped if lost. No retransmission, no stalling. This is ideal for data that becomes stale quickly—game state updates, real-time cursor positions, live sensor readings—where receiving the *latest* value matters more than receiving *every* value.

- **Multiple independent reliable streams.** An application can open many parallel streams, each with guaranteed in-order delivery, but without head-of-line blocking between them. A stall on one stream does not affect the others. This is useful for sending different types of data (e.g., chat messages, control signals, file transfers) simultaneously without interference.
- **Connection migration.** Inherited from QUIC: the connection survives network changes such as Wi-Fi to cellular handoffs.

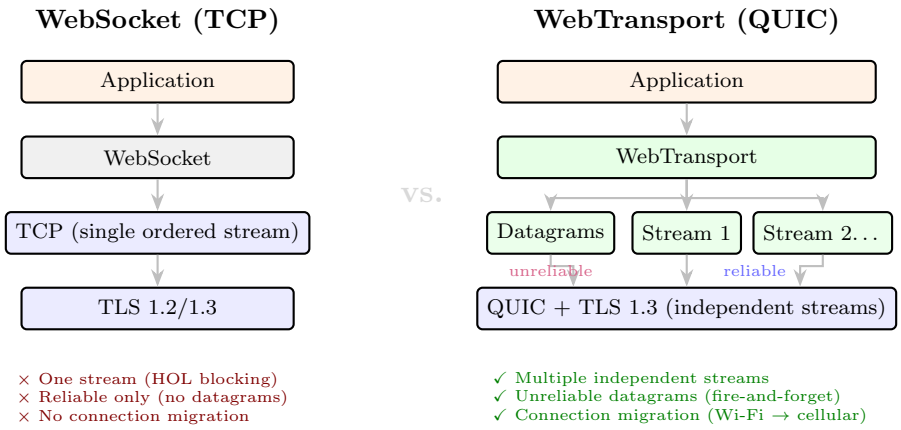


Figure 5: WebSocket vs. WebTransport. WebSocket provides a single ordered stream over TCP; a lost packet stalls everything. WebTransport provides multiple independent streams *and* unreliable datagrams over QUIC, with connection migration.

WebTransport is a *client-server* protocol: the browser connects to a server, not directly to another browser. It complements WebRTC—where WebRTC provides direct browser-to-browser media exchange, WebTransport provides a modern browser-to-server data path suited to relayed media architectures, game state synchronization, and real-time telemetry.

2.6 WebCodecs: Direct Codec Access

WebCodecs (W3C Working Draft) gives JavaScript code direct, frame-by-frame access to the audio and video codecs built into the browser.

To understand why this matters, consider how browsers have traditionally handled media encoding and decoding. High-level APIs like `MediaRecorder` (for recording) and `HTMLMediaElement` (for playback) control the entire pipeline: the browser decides when to produce keyframes, how to adjust bitrate, and what container format to use. The application gets a finished recording or a rendered video—but cannot intervene at the level of individual frames.

WebCodecs breaks open this pipeline. It provides four interfaces:

- `VideoEncoder` / `VideoDecoder` — encode or decode individual video frames, with per-frame control over keyframe insertion, target bitrate, and quality parameters.
- `AudioEncoder` / `AudioDecoder` — the same for individual audio chunks.

These interfaces use the browser's built-in hardware-accelerated codec implementations. When dedicated encoding hardware is available (as on most modern GPUs and mobile chips), WebCodecs uses it automatically; otherwise it falls back to software encoding. The application gets raw encoded bytes without container overhead, and can process, transform, or transport them however it chooses.

WebCodecs is deliberately *transport-agnostic*—it handles encoding and decoding, not delivery. The encoded output can be sent via WebRTC (using Encoded Transforms to insert custom processing into the `RTCPeerConnection` pipeline), via WebTransport (packaging encoded frames as datagrams or stream data), or via any other mechanism.

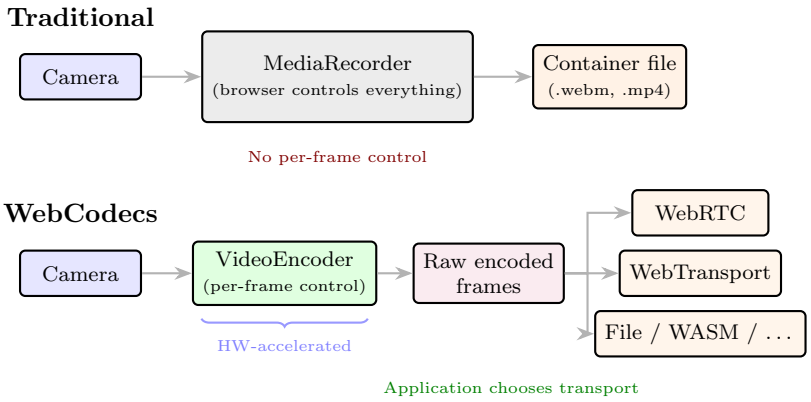


Figure 6: Traditional codec access vs. WebCodecs. Top: `MediaRecorder` controls the entire pipeline; the application receives a finished container file. Bottom: WebCodecs gives per-frame control over hardware-accelerated encoding, and the application decides how to transport the raw output.

2.7 How the Standards Interact

The standards described above can be summarized in four key relationships:

1. **WebRTC is self-contained for peer-to-peer communication.** A single API call, `RTCPeerConnection`, bundles ICE (path discovery), DTLS-SRTP (encryption), RTP (media transport), and codec negotiation. An application can build a complete video call using only WebRTC and an application-provided signaling channel, without touching any other standard discussed here.

2. **QUIC → HTTP/3 → WebTransport is a layered client-server path.** Each layer adds capability: QUIC provides multiplexed encrypted transport over UDP, HTTP/3 maps HTTP semantics onto it, and WebTransport exposes QUIC's streams and datagrams to JavaScript. An application that needs browser-to-server real-time communication uses all three layers, but interacts only with the WebTransport API at the top.

3. **WebCodecs bridges both paths.** In a WebRTC context, WebCodecs (via Encoded Transforms) enables custom processing of media frames flowing through `RTCPeerConnection`—for example, applying end-to-end encryption inside the WebRTC pipeline. In a WebTransport context, WebCodecs encodes and decodes media that the application transports itself, independent of WebRTC.

4. **The "unbundled" alternative.** Rather than using WebRTC's monolithic `RTCPeerConnection`,

an application can compose WebTransport (for transport) + WebCodecs (for encoding/decoding) + WebAssembly (for custom processing) to build a fully custom media pipeline. This trades WebRTC's built-in integration for explicit control over every decision: which codec, what packetization, how to handle congestion, what to retransmit. This architecture is projected to become viable by 2029 (Sections 8–9).

3 WebRTC

The WebRTC stack in the `standards/` directory comprises the W3C WebRTC 1.0 specification and six IETF RFCs: JSEP (RFC 8829), ICE (RFC 8445), the WebRTC security considerations (RFC 8826), the DTLS-based security architecture (RFC 8827), RTP (RFC 3550), and SRTP (RFC 3711).

3.1 Late 2021

W3C WebRTC 1.0 achieved Recommendation status in January 2021, making it a finalized standard. By late 2021, Chrome and Firefox offered full implementations, and Edge inherited Chrome's support via its Chromium base. Safari supported the core API but with significant limitations:

- **Codec restrictions.** Safari supported only H.264 for video. Chrome and Firefox supported VP8, VP9, and H.264. This meant Safari could not participate in VP9-based calls and lacked access to VP9's more efficient bandwidth usage.
- **No simulcast or SVC.** Safari could not send multiple encodings of the same video stream at different resolutions (simulcast) or use Scalable Video Coding (SVC). Chrome and Firefox supported both, enabling adaptive quality based on network conditions.
- **RTCDataChannel issues on iOS.** The data channel did not work reliably on iOS Safari. Cross-browser data channel connections between Safari on iOS and Chrome or Firefox failed due to missing local ICE candidates on iOS.
- **SDP handling divergence.** Chrome was in the process of migrating from Plan B to Unified Plan SDP semantics. Chrome M89 (February 2021) added deprecation warnings; Chrome M93 (August 2021) removed Plan B with a reverse origin trial option. Firefox and Safari already used Unified Plan exclusively. During this transition, applications had to accommodate both SDP formats.
- **Insertable Streams.** Chrome M94 introduced the Insertable Streams API, enabling JavaScript to access and transform encoded media frames in the WebRTC pipeline. This was essential for end-to-end encryption of media. Firefox and Safari did not implement it.
- **Stats reporting inconsistencies.** Firefox was missing `voiceActivityFlag` on video track stats. Safari was missing `rtpTimestamp` and `voiceActivityFlag`. Applications could not reliably detect voice activity or obtain precise timing information across browsers.

On the IETF side, the supporting protocols were in the following state:

- **JSEP (RFC 8829)** was published in January 2021 alongside the W3C Recommendation. All browsers implemented the JSEP signaling state machine through `RTCPeerConnection`, but diverged in areas like BUNDLE handling of multiple media streams.

- **ICE (RFC 8445)** was implemented in all browsers. Trickle ICE (RFC 8838), enabling incremental candidate exchange rather than waiting for full gathering, was published in January 2021 but not yet mandatory.
- **DTLS-SRTP (RFCs 8827, 3711)** was mandatory across all browsers. All media was encrypted. However, browsers differed in DTLS certificate algorithms (Chrome 52+ used ECDSA; others varied), and privacy identifier reset mechanisms were inconsistent.
- **Security (RFC 8826)** — all browsers enforced camera/microphone permission prompts and encrypted all media. However, protections against IP address leakage were inconsistent; not all browsers masked local IP addresses via mDNS candidates when media permissions had not been granted.

3.2 March 2026

The W3C updated the WebRTC 1.0 Recommendation in October 2024 and again in 2025, incorporating amendments based on implementation experience.

- **Unified Plan is universal.** Plan B SDP is completely removed from all browsers. Applications no longer need to accommodate both formats.
- **Safari gained VP9.** Safari 15 and later support the VP9 codec for WebRTC, enabling Safari to participate in VP9-based calls and benefit from VP9's bandwidth efficiency. Safari still prefers H.264 for hardware acceleration, and its SVC support remains limited compared to Chrome and Firefox.
- **iOS WebRTC matured.** Starting with iOS 14.3, WebRTC works reliably across all browsers on iOS. RTCDataChannel connections between iOS Safari and desktop Chrome or Firefox are reliable.
- **JSEP updated to RFC 9429.** Published in April 2024, RFC 9429 obsoleted RFC 8829, addressing implementation divergence — particularly in BUNDLE handling and bundle-only `m=` sections.
- **Trickle ICE is mandatory.** All browsers implement RFC 8838, enabling faster connection establishment through incremental candidate exchange.
- **mDNS privacy protections standardized.** Chrome, Firefox, and Safari all mask local IP addresses using mDNS candidates when camera/microphone permissions have not been granted, implementing the privacy considerations of RFC 8826.
- **Encoded Transforms remain Chrome-only.** The feature evolved from "Insertable Streams" to `RTCRtpScriptTransform` (WebRTC Encoded Transform), but Firefox and Safari still do not implement it. End-to-end encryption of encoded media frames in the WebRTC pipeline remains a Chrome/Chromium-only capability.

4 QUIC

RFC 9000 defines the QUIC transport protocol: a UDP-based, multiplexed, encrypted transport with built-in TLS 1.3, independent stream flow control, and connection migration across network changes.

4.1 Late 2021

RFC 9000 was published in May 2021. Browser support was uneven:

- **Chrome** had QUIC enabled by default, having been an early adopter (Google developed the original QUIC protocol).
- **Firefox 88** (April 2021) enabled QUIC by default.
- **Safari** had experimental QUIC support (added in Safari Technology Preview 104, April 2020), but it required manual opt-in through experimental settings. It was not available to ordinary users.

In practice, a browser connecting to a QUIC-capable server would benefit from reduced connection latency (0-RTT or 1-RTT handshakes vs. TCP's multi-round handshake plus TLS), elimination of head-of-line blocking across multiplexed streams, and connection survival across network changes. But with Safari requiring manual opt-in, these benefits were unavailable to most Safari users.

4.2 March 2026

QUIC is enabled by default in all major browsers:

- Chrome, Edge, and Firefox continue default support.
- Safari 16 (2022) enabled QUIC by default, bringing universal coverage.

Every major browser can now establish QUIC connections transparently. The capabilities defined by RFC 9000 — low-latency connection establishment, multiplexed streams without head-of-line blocking, built-in encryption, and connection migration — are universally available to web applications without any user or developer opt-in.

5 HTTP/3

RFC 9114 maps HTTP semantics over QUIC, inheriting QUIC's transport properties: multiplexed streams without head-of-line blocking, lower connection latency, and connection migration.

5.1 Late 2021

RFC 9114 had not yet been published (it was finalized in June 2022). Browsers were shipping pre-standard HTTP/3 implementations based on draft specifications:

- **Chrome** had experimental HTTP/3 support enabled by default since April 2020.
- **Firefox 88** (April 2021) enabled HTTP/3 by default.
- **Safari** had HTTP/3 support behind an experimental feature flag, disabled by default.

Where supported, browsers could negotiate HTTP/3 with compatible servers, gaining per-stream independence (a stalled stream would not block other streams, unlike HTTP/2 over TCP). However, the lack of a finalized RFC meant implementations were based on draft specifications that could diverge, and Safari's experimental-only status limited universal availability.

5.2 March 2026

With RFC 9114 finalized in June 2022, all major browsers now support HTTP/3 by default:

- Chrome, Edge, and Firefox have supported HTTP/3 by default for several years.
- Safari enabled HTTP/3 by default starting with Safari 16 (2022).
- Over 92% of browser market share supports HTTP/3.

The capabilities defined by RFC 9114 — HTTP request/response multiplexing without head-of-line blocking, lower latency connection establishment via QUIC, and connection resilience across network changes — are now universally available. Any browser connecting to an HTTP/3-capable server automatically benefits from these properties.

6 WebTransport

The WebTransport specification defines browser APIs for bidirectional client-server communication over HTTP/3, supporting both reliable streams and unreliable datagrams. RFC 9297 defines the HTTP datagram and capsule protocol mechanisms that underpin it.

6.1 Late 2021

WebTransport was in a very early state:

- **Chrome** had experimental support but it was disabled in Chrome M91 through M94. The API was not usable in production.
- **Firefox** had no support whatsoever.
- **Safari** had no support whatsoever.
- RFC 9297 had not yet been published (August 2022).
- The W3C specification was an early Working Draft.

In late 2021, WebTransport was effectively unavailable in any browser for production use. The capabilities it defines — unreliable datagrams from browser to server, multiple independent reliable streams, and connection migration — could not be used by web applications.

6.2 March 2026

WebTransport has achieved broad but not universal support:

- **Chrome 97+** and **Edge 98+** support WebTransport (stable since late 2021 / early 2022).
- **Firefox 114+** (2023) supports WebTransport.
- **Safari does not support WebTransport.**
- Approximately 82% of browser market share has WebTransport support.

For browsers that support it, WebTransport enables capabilities that were previously impossible without WebRTC's peer-to-peer architecture:

- **Unreliable datagrams** from browser to server, suitable for latency-sensitive data where retransmission is undesirable (e.g., game state updates, real-time sensor data).
- **Multiple independent reliable streams** without head-of-line blocking between them, enabling parallel data flows over a single connection.
- **Connection migration** across network changes, inherited from QUIC.

Safari's absence means applications using WebTransport must provide fallback mechanisms (typically WebSocket) for Safari users, or accept that these capabilities are unavailable on Apple platforms.

7 WebCodecs

The WebCodecs specification provides JavaScript interfaces for direct access to the browser's built-in audio, video, and image codecs, enabling frame-level encoding and decoding outside of higher-level APIs like `MediaRecorder` or `HTMLMediaElement`.

7.1 Late 2021

WebCodecs had just reached initial availability:

- **Chrome 94** (September 2021) shipped WebCodecs as a stable feature, graduating from origin trial. Chrome 92 and 93 had offered it as an origin trial with breaking changes between versions.
- **Firefox** had no support.
- **Safari** had no support.
- The W3C specification was a First Public Working Draft (2021).

In Chrome 94, developers could for the first time:

- Encode and decode individual video frames using `VideoEncoder` and `VideoDecoder` with the browser's hardware-accelerated codecs, without being constrained by `MediaRecorder`'s real-time recording model.
- Encode and decode individual audio chunks (`AudioEncoder`/`AudioDecoder`).
- Perform faster-than-realtime encoding (previously impossible via `MediaRecorder`).
- Control per-frame keyframe insertion and quality parameters.
- Obtain raw encoded bytes without container overhead.

However, these capabilities were Chrome-only. Firefox and Safari users had no access to low-level codec APIs; they were limited to the high-level `MediaRecorder` and `HTMLMediaElement` interfaces.

7.2 March 2026

WebCodecs has achieved broad cross-browser support:

- **Chrome/Edge** — supported since version 94 (2021).
- **Firefox 130+** (September 2024) — shipped approximately three years after Chrome.

- **Safari 26+** (2024) — full support. Earlier Safari versions (16.4+) had partial video-only support.
- Approximately 94% of browser market share supports WebCodecs.
- The specification remains a W3C Working Draft (updated January 2026), not yet a Recommendation.

With cross-browser support, all major browsers can now perform frame-level codec operations using hardware acceleration. This enables web applications to build custom media pipelines — video editors, live transcoders, custom streaming clients — that were previously possible only through WebAssembly codec implementations or native applications.

8 Looking Ahead: 2028–2029 (Two to Three Years)

The following projections are grounded in active W3C charter milestones, IETF draft publication timelines, and formal browser vendor implementation signals.

8.1 WebCodecs Reaches W3C Recommendation

The W3C Media Working Group charter targets WebCodecs Candidate Recommendation in Q1 2026 and full Recommendation by Q4 2026. By 2028, the specification should be a mature, finalized standard. This means the frame-level codec access that was Chrome-only in 2021 and cross-browser by 2026 will have a stable, locked-down API surface that browser vendors commit to maintaining long-term. The WebCodecs Codec Registry will likely include formal AV1 registration entries alongside existing H.264, VP8, and VP9 entries.

8.2 WebTransport Achieves Universal Browser Support

Apple committed WebTransport as a formal Interop 2026 priority, accounting for 20% of the interoperability score. Safari already has experimental WebTransport support behind a flag in iOS 18. Given this commitment, Safari production support is expected by late 2026 or early 2027, with the W3C specification reaching Recommendation status around the same time. By 2028, WebTransport should be universally available across all major browsers — closing the last significant gap in the stack. Applications will be able to rely on browser-to-server unreliable datagrams and independent reliable streams without Safari fallbacks.

8.3 WebRTC Encoded Transforms Become Cross-Browser

The `RTCRtpScriptTransform` API (WebRTC Encoded Transform) has a Working Draft dated February 2026 and is approaching Candidate Recommendation. Safari and Firefox have implemented it natively; Chrome continues support via its earlier Insertable Streams lineage. By 2028, this API should be standardized and available across all major browsers, enabling cross-browser end-to-end encryption of media frames within the WebRTC pipeline — a capability that was Chrome-only from 2021 through 2026.

8.4 AV1 Becomes a Practical WebRTC Codec

Firefox enabled AV1 for WebRTC by default in 2025. Chrome has supported it for several years. By 2028, AV1 hardware encoding support in consumer devices will have expanded, reducing the

CPU cost that currently limits its adoption. AV1's superior compression efficiency (roughly 30% better than VP9 at equivalent quality) will make it a practical default codec for WebRTC calls in Chrome and Firefox. Safari's adoption remains uncertain, as Apple historically favors H.264 with hardware acceleration.

8.5 Media over QUIC Reaches RFC Status

The IETF Media over QUIC (MoQ) transport draft (`draft-ietf-moq-transport`) is advancing rapidly, with version 15 published in October 2025. The protocol enables subscription-based media publishing and consuming over QUIC, designed for low-latency distribution with CDN support. By 2028, MoQ should have reached RFC status, along with companion specifications for media container formats (Low Overhead Media Container) and streaming formats. Browser integration will likely occur through JavaScript libraries built on WebTransport and WebCodecs rather than through dedicated browser APIs — the component APIs are already in place to support this.

8.6 RTP over QUIC Reaches RFC Status

The IETF RTP over QUIC (RoQ) draft (`draft-ietf-avtcore-rtp-over-quic`) maps RTP and RTCP packets onto QUIC streams and datagrams. By 2028, this should reach RFC status, enabling WebRTC backends to transport media over QUIC instead of the traditional UDP+DTLS-SRTP path. Browser-facing integration would occur through SDP negotiation rather than new JavaScript APIs, potentially enabling WebRTC calls to benefit from QUIC's congestion control and connection migration at the transport layer.

8.7 Hardware and Network Context

The standards projections above do not exist in a vacuum. By 2028, two underlying trends amplify what these standards make practically possible.

CPU and hardware acceleration. AV1 encoding is computationally expensive — in 2026, real-time AV1 encoding for a WebRTC call can consume an entire CPU core on mid-range hardware, which is why VP9 and H.264 remain the practical defaults despite AV1's superior compression. By 2028, dedicated AV1 hardware encoders will be standard in new laptops, phones, and tablets (Apple's M-series chips, Qualcomm Snapdragon, Intel Arc, and AMD RDNA already include or are adding AV1 encode blocks). This shifts AV1 from "technically supported but too expensive" to "hardware-accelerated and free." The WebCodecs API, which exposes hardware-accelerated codecs to JavaScript, directly benefits: an application calling `VideoEncoder` with AV1 in 2028 will hit dedicated silicon rather than a software encoder, making frame-level AV1 processing via WebCodecs practical for real-time use cases.

Similarly, the unbundled architecture — composing WebTransport, WebCodecs, and WebAssembly — imposes CPU overhead from JavaScript orchestration and WebAssembly processing that would have been prohibitive on 2021 hardware. By 2028, the roughly 40–50% improvement in single-thread CPU performance since 2021 (compounding at approximately 8–10% annually) makes this overhead manageable for real-time media pipelines.

Network bandwidth. Global median fixed broadband speeds have approximately doubled from 2021 to 2026 (from roughly 60 Mbps to over 120 Mbps in developed markets), and 5G mobile networks have expanded substantially. By 2028, continued growth in fixed and mobile bandwidth means that the efficiency gains from AV1 and QUIC compound with raw bandwidth increases. A

WebRTC call using AV1 over QUIC on a 2028 connection will have both a more efficient codec (30% less bandwidth than VP9) and more available bandwidth, enabling higher resolution and frame rates within the same network cost envelope. QUIC's connection migration (RFC 9000) becomes more impactful as mobile usage grows — a video call that seamlessly survives a Wi-Fi to 5G handoff requires both the protocol support (already standardized) and sufficient mobile bandwidth to sustain the call (increasingly available).

The net effect is that the standards reaching maturity by 2028 will land on hardware and networks that can fully exploit them, rather than being bottlenecked by the infrastructure they run on.

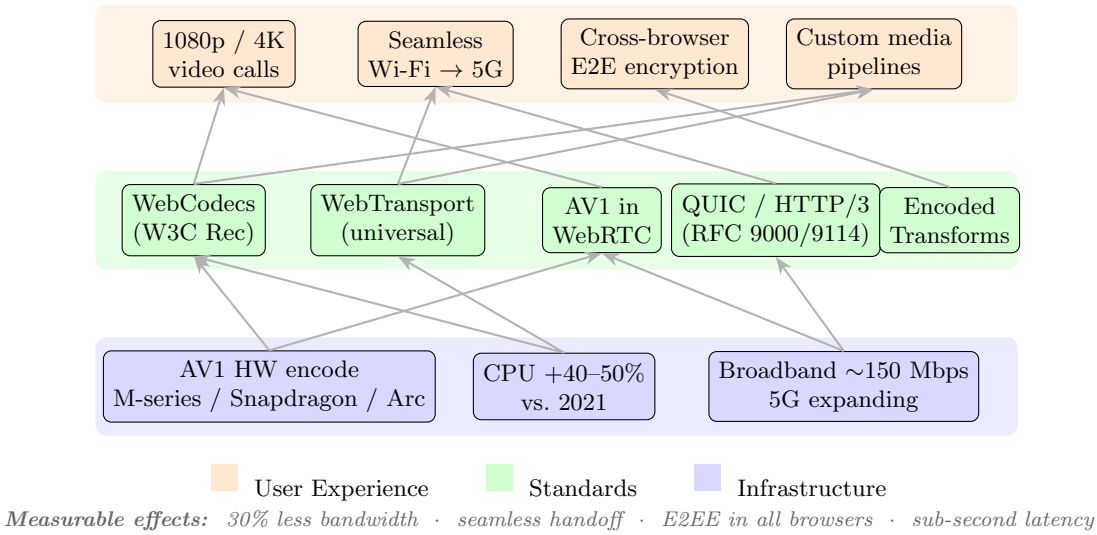


Figure 7: By 2028: infrastructure enables standards to deliver measurable user experience gains. Arrows show enabling relationships.

9 Looking Ahead: 2029–2031 (Three to Five Years)

These projections extend further from current standardization activity and carry correspondingly more uncertainty, but follow from the trajectory established by the standards work described above.

9.1 The "Unbundled" Architecture Becomes Viable

By 2029, all the component APIs — WebTransport (universal), WebCodecs (Recommendation), and RTCRtpScriptTransform (standardized) — will have been stable and cross-browser for two or more years. This makes the "WebRTC Unbundled" architecture practically viable: applications can compose WebTransport for data transport, WebCodecs for encoding and decoding, and WebAssembly for custom processing, bypassing the monolithic `RTCPeerConnection` API entirely.

This matters because the unbundled approach gives applications control over decisions that WebRTC currently makes internally: which codec to use, how to packetize, what transport to use, and how to handle congestion. Use cases that are poorly served by WebRTC's built-in assumptions — cloud gaming requiring ultra-low latency with acceptable loss, scientific visualization requiring custom frame processing, or broadcast distribution requiring one-to-many fan-out — can be built from these composable primitives.

Traditional WebRTC via `RTCPeerConnection` will persist for general video conferencing, where its built-in codec negotiation, ICE connectivity, and DTLS-SRTP security provide a well-integrated default. The unbundled approach is not a replacement but an alternative for applications with specialized requirements.

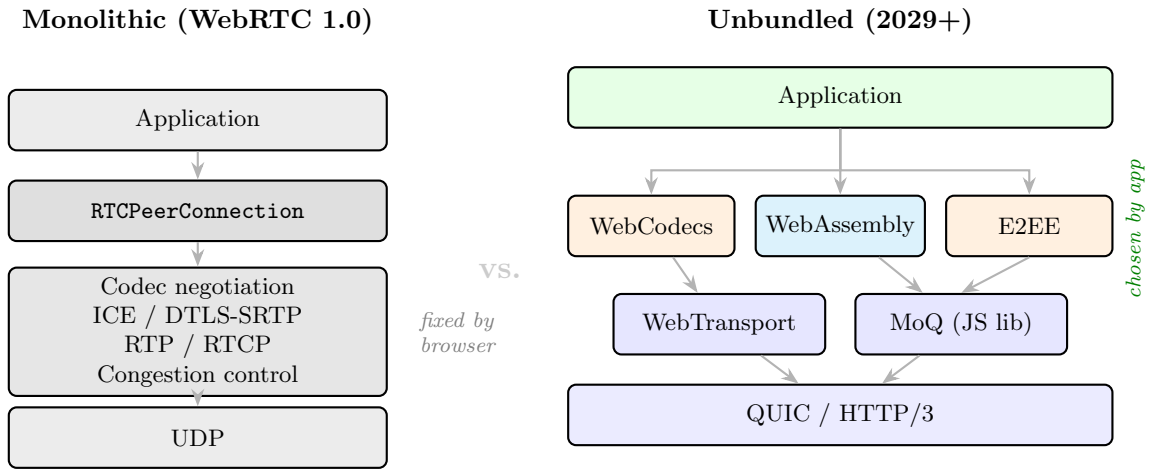


Figure 8: Monolithic WebRTC (left) vs. the unbundled architecture (right). The unbundled approach gives applications control over codec, transport, encryption, and congestion decisions that `RTCPeerConnection` handles internally.

9.2 MoQ Enables Standards-Based Low-Latency Streaming

With MoQ at RFC status and WebTransport universally available, a standards-based path exists for low-latency media streaming from server to browser that does not require WebRTC. A MoQ JavaScript library running on WebTransport and using WebCodecs for decoding could deliver sub-second latency streaming with CDN fan-out — combining the latency characteristics previously unique to WebRTC with the scalability of traditional CDN architectures. This creates a standards-based alternative to proprietary low-latency streaming protocols.

9.3 AV1 as the Default WebRTC Codec

By 2030, hardware AV1 encoding will be standard in consumer devices (laptops, phones, tablets). The CPU cost barrier that limits AV1 adoption in 2026 will have largely disappeared. AV1's compression advantage over VP9 and H.264 makes it the natural default for WebRTC, reducing bandwidth requirements for equivalent video quality. Whether Safari adopts AV1 for WebRTC depends on Apple's hardware acceleration decisions, but the broader ecosystem pressure from Chrome and Firefox — which together represent the majority of browser usage — will make AV1 the dominant codec for new WebRTC applications.

9.4 QUIC v2 Transparent Adoption

QUIC v2 (RFC 9369, published May 2023) is identical to QUIC v1 in capability but uses different wire constants to resist middlebox ossification. By 2030, browsers will have transparently migrated to QUIC v2 for new connections. This is invisible to applications — no new capabilities are exposed — but ensures the QUIC transport layer remains resilient against network intermediaries that might otherwise interfere with the protocol based on recognizing v1 byte patterns.

9.5 No WebRTC 2.0

The W3C WebRTC Working Group charter (extending to September 2026) adds new capabilities through extension specifications rather than monolithic version bumps. This pattern is expected to continue. There will be no "WebRTC 2.0" in the traditional sense. Instead, the API surface will grow incrementally through standardized extensions — Encoded Transforms, SFrame for end-to-end encryption in group calls, and future extensions not yet drafted — while the core `RTCPeerConnection` API remains stable.

9.6 Hardware and Network Context

By 2030, the hardware and network landscape further amplifies the practical impact of these matured standards.

CPU and hardware acceleration. By 2030, single-thread CPU performance will have improved roughly 60–80% relative to 2021 levels, and AV1 hardware encode/decode will be ubiquitous — present not just in flagship devices but in budget laptops and mid-range phones. This has two consequences for the standards discussed here. First, AV1 becomes the natural default codec for WebRTC: the hardware acceleration eliminates the CPU cost barrier, and WebCodecs applications calling `VideoEncoder` with AV1 will consistently hit dedicated silicon across the device spectrum, not just on high-end hardware. Second, the unbundled architecture — WebTransport plus WebCodecs plus WebAssembly — becomes practical even on resource-constrained devices. The JavaScript and WebAssembly orchestration overhead that would strain a 2021 mobile processor is routine on a 2030 device, making custom real-time media pipelines viable on phones and tablets, not just desktop machines.

Network bandwidth. If current trends hold, median fixed broadband speeds in developed markets will reach 200–300 Mbps by 2030, with 5G mobile networks providing reliable 100+ Mbps in urban areas. At these speeds, bandwidth is no longer the binding constraint for most real-time communication scenarios. The practical consequence is a shift in what matters: codec efficiency (AV1 over H.264) becomes less about fitting within tight bandwidth limits and more about enabling higher quality — 4K video conferencing, high-frame-rate screen sharing, or multi-stream layouts — without saturating the connection. QUIC's multiplexed streams (RFC 9000) and HTTP/3's head-of-line blocking elimination (RFC 9114) matter most when multiple parallel flows compete for bandwidth, which is precisely the scenario enabled by abundant bandwidth and ambitious applications.

MoQ-based streaming benefits particularly from this environment. A MoQ library on WebTransport using WebCodecs to decode AV1 can deliver high-bitrate, low-latency streams that would have been impractical in 2021 not because the standards didn't exist, but because the combination of codec cost, network capacity, and API availability wasn't there. By 2030, all three constraints are resolved simultaneously: the standards are finalized and universal, the hardware accelerates the codecs, and the networks carry the bits.

The compounding effect. The key observation is that standards maturation, hardware improvement, and network growth are complementary, not independent. A standard that reaches browser universality on hardware that can't run it efficiently, or over networks that can't carry its output, delivers theoretical capability rather than practical impact. By the 2029–2031 timeframe, these three curves converge: every standard in this document will be finalized and universally implemented, running on hardware with dedicated acceleration for the codecs they specify, over networks with

bandwidth to spare. This convergence is what makes the 3–5 year outlook qualitatively different from a simple extrapolation of today's capabilities.

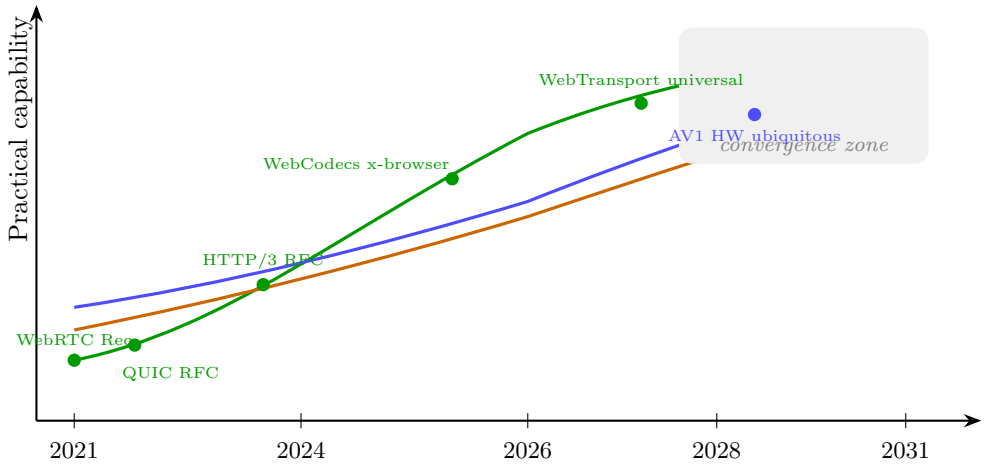


Figure 9: Convergence of standards maturity, hardware capability, and network bandwidth from 2021 to 2031. The shaded region marks where all three curves reach levels sufficient to fully exploit the standards discussed in this document.

10 Summary

11 Conclusion

This document has traced the real-time web communication stack across a full decade, from its fragmented state in late 2021 through its current maturity in March 2026 to its projected convergence by 2031.

In late 2021, the stack was transitional. WebRTC was the only finalized standard with broad browser support, yet even it suffered from significant cross-browser inconsistencies — Safari's codec and feature limitations, Chrome's ongoing SDP migration, and iOS reliability issues. QUIC had just been standardized but was not yet universally enabled. HTTP/3 was shipping based on draft specifications. WebTransport was effectively non-functional in any browser. WebCodecs had just shipped in a single browser. MoQ did not yet exist as a draft.

By March 2026, the picture is fundamentally different. The core protocol stack — from QUIC transport through HTTP/3 to WebRTC — is standardized via finalized RFCs and implemented by default across all major browsers. WebCodecs provides cross-browser access to hardware-accelerated codecs at the frame level. Two gaps remain: Safari's lack of WebTransport support and the Chrome-only status of Encoded Transforms for end-to-end encryption.

The 2028–2029 projections close these gaps. WebTransport reaches Safari via Apple's Interop 2026 commitment. Encoded Transforms become cross-browser. WebCodecs achieves W3C Recommendation status. AV1 becomes a practical WebRTC codec as hardware encoding expands. MoQ and RTP over QUIC reach RFC status, establishing new standards-based paths for media transport. Hardware and network improvements — 40–50% CPU gains since 2021, broadband speeds approaching 150 Mbps — ensure these standards land on infrastructure that can fully exploit them.

Standard	Late 2021	March 2026	2028–2031 Projection
WebRTC 1.0	W3C Rec (Jan 2021). Chrome/Firefox full support. Safari limited: H.264 only, no simulcast, iOS DataChannel broken. Plan B still in Chrome.	Updated Rec (2024–25). Unified Plan universal. Safari has VP9. iOS reliable. JSEP updated to RFC 9429. Encoded Transforms Chrome-only.	Encoded Transforms cross-browser. AV1 default codec (HW-accelerated). SFrame for group E2EE. No "WebRTC 2.0" — incremental extensions.
QUIC	RFC 9000 (May 2021). Chrome/Firefox default. Safari experimental, manual opt-in.	Default in all browsers (Safari 16+, 2022).	QUIC v2 (RFC 9369) transparent migration. RTP over QUIC (RoQ) at RFC status.
HTTP/3	Pre-RFC draft implementations. Chrome/Firefox default. Safari behind flag.	RFC 9114 (June 2022). Default in all browsers. 92%+ coverage.	Mature universal transport. Foundation for WebTransport and MoQ.
WebTransport	Experimental only. Chrome disabled (M91–94). Firefox/Safari: none.	Chrome 97+, Firefox 114+. Safari: still absent. 82% coverage.	Universal (Safari via Interop 2026). W3C Rec. Enables unbundled architecture and MoQ.
WebCodecs	Chrome 94 only (Sept 2021). Firefox/Safari: none. First Public WD.	Chrome, Firefox 130+, Safari 26+. 94% coverage. Still WD.	W3C Rec by 2027. AV1 HW encode ubiquitous. Custom media pipelines on all devices.
MoQ	Did not exist.	IETF draft (v15, Oct 2025). No browser API.	RFC status. JS libraries on WebTransport + WebCodecs. Sub-second CDN streaming.

Table 1: Browser capability status by standard across the full 2021–2031 timeline.

By 2029–2031, the stack reaches full convergence. The "unbundled" architecture — composing WebTransport, WebCodecs, and WebAssembly as alternatives to monolithic `RTCPeerConnection` — becomes practically viable for specialized use cases like cloud gaming, scientific visualization, and broadcast distribution. AV1 hardware encoding is ubiquitous across the device spectrum. MoQ on WebTransport enables standards-based sub-second streaming with CDN fan-out. Network bandwidth (200–300 Mbps fixed, 100+ Mbps mobile) shifts the constraint from capacity to quality: 4K conferencing, multi-stream layouts, and high-frame-rate screen sharing become routine.

The central observation is that standards maturation, hardware improvement, and network growth are complementary, not independent. A standard that reaches browser universality on hardware that cannot run it efficiently, or over networks that cannot carry its output, delivers theoretical capability rather than practical impact. The 2029–2031 timeframe is where all three curves converge: every standard in this document will be finalized and universally implemented, running on hardware with dedicated codec acceleration, over networks with bandwidth to spare. This convergence is what distinguishes the projected end state from a simple extrapolation of today's capabilities — and what makes the full real-time web communication stack, from transport to codec to application, a resolved problem rather than a work in progress.

A Information-Theoretic View

This appendix applies information-theoretic measures to the real-time web communication stack described in this document. We quantify channel capacity, source coding efficiency, and effective information rates across three time horizons: today (March 2026), the near term (2027–2028), and the medium term (2029–2031). The analysis draws on the standards, codec parameters, and infrastructure projections developed in the main text.

A.1 Channel Capacity of the Transport Layer

The Shannon-Hartley theorem gives the theoretical maximum information rate for a channel with additive white Gaussian noise:

$$C = B \log_2\left(1 + \frac{S}{N}\right) \quad [\text{bits/s}] \tag{1}$$

where B is the channel bandwidth in Hz and S/N is the signal-to-noise ratio.

While the physical-layer capacity of a broadband link is governed by (**??**), the *effective* capacity available to the application is reduced by protocol overhead, congestion control, and transport inefficiencies. We define the effective application-layer throughput as:

$$R_{\text{eff}} = C_{\text{link}} \cdot \eta_{\text{transport}} \cdot (1 - \rho_{\text{overhead}}) \tag{2}$$

where C_{link} is the link-layer capacity, $\eta_{\text{transport}} \in (0, 1]$ is the transport efficiency factor (capturing congestion control, retransmission, and multiplexing behavior), and $\rho_{\text{overhead}} \in [0, 1)$ is the fractional protocol overhead (headers, encryption, framing).

A.1.1 Head-of-Line Blocking and Multiplexed Capacity

Consider n independent streams multiplexed over a single connection. Under TCP (HTTP/2), a single lost packet blocks all n streams. The expected throughput per stream under loss rate p is bounded by:

$$R^{(i)}_{\text{TCP}} \leq \frac{R_{\text{eff}}}{n} \cdot (1 - p)^{n-1} \tag{3}$$

because a loss in *any* of the n streams stalls all others. Under QUIC (RFC 9000), streams are independent at the transport layer, so:

$$R^{(i)}_{\text{QUIC}} \leq \frac{R_{\text{eff}}}{n} \cdot (1 - p) \tag{4}$$

The ratio of effective per-stream throughput is:

$$\frac{R^{(i)}_{\text{QUIC}}}{R^{(i)}_{\text{TCP}}} = \frac{1}{(1 - p)^{n-2}} \tag{5}$$

Excursus: Why $n = 8$? — In a typical multi-participant video conference, each peer contributes multiple concurrent streams: a camera video track, an audio track, and often a screen-share or data channel. A four-person call with two media tracks each produces $n = 8$ multiplexed streams over a single connection — the reference value used throughout this section. Larger meetings (e.g. 10 participants with simulcast layers) push n higher, amplifying the HOL blocking effect further.

For $n = 8$ streams (a typical multi-track video conference) at $p = 0.01$ (1% loss), this gives a QUIC advantage factor of $1/(0.99)^6 \approx 1.062$, a modest 6.2% improvement. At $p = 0.05$ (5% loss, common on mobile or congested links), the factor is $1/(0.95)^6 \approx 1.34$, a 34% throughput advantage per stream.

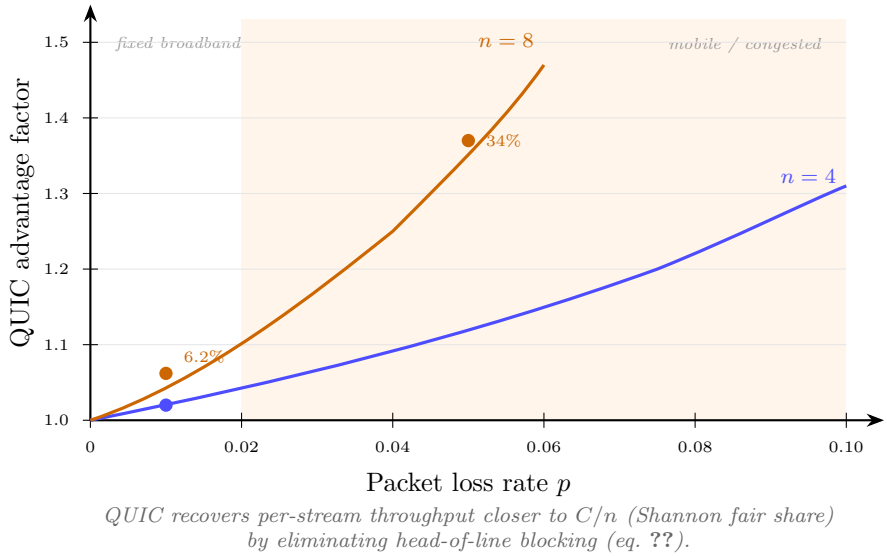


Figure 10: QUIC advantage factor $1/(1-p)^{n-2}$ over TCP for multiplexed streams. At 5% loss with $n = 8$ streams, QUIC delivers 34% more per-stream throughput by maintaining independence at the transport layer.

A.1.2 Connection Establishment Latency

Let τ denote the one-way latency (half the round-trip time). The information-carrying capacity during connection establishment is zero; the handshake represents pure overhead. The time to first useful byte is:

$$T_{\text{TCP+TLS 1.3}} = 2\tau_{\text{TCP}} + \tau_{\text{TLS}} = 3\tau \quad \text{(1-RTT TLS 1.3)} \tag{6}$$

$$T_{\text{QUIC}} = \tau \quad \text{(1-RTT)} \tag{7}$$

$$T_{\text{QUIC, 0-RTT}} = 0 \quad \text{(resumption)} \tag{8}$$

The QUIC 0-RTT case (**??**) eliminates the handshake entirely for resumed connections. The information lost during handshake dead time, relative to steady-state capacity, is:

$$I_{\text{lost}} = C_{\text{link}} \cdot T_{\text{setup}} \tag{9}$$

For a 120 Mbps link with $\tau = 25$ ms (typical fixed broadband in 2026), the TCP+TLS 1.3 handshake wastes $120 \times 10^6 \times 0.075 = 9 \times 10^6$ bits (1.125 MB), while QUIC 1-RTT wastes only 3×10^6 bits (375 KB).

A.2 Source Coding Efficiency

The rate-distortion function $R(D)$ defines the minimum bit rate required to represent a source at distortion level D. For a Gaussian source with variance σ^2 under mean squared error distortion:

$$R(D) = \frac{1}{2}\log_2 \frac{\sigma^2}{D} \quad \text{[bits/sample]} \tag{10}$$

Excursus: Why Gaussian source? — The Gaussian memoryless model is used because it yields the highest (worst-case) $R(D)$ among all sources of equal variance. Any codec that approaches the Gaussian bound on real video is therefore doing at least as well as the bound predicts — the true $R(D)$ for spatially and temporally correlated video lies below the Gaussian curve, making it a conservative benchmark.

No practical codec achieves $R(D)$;[1] real codecs operate at some rate $R_{\text{codec}} > R(D)$. We define the coding efficiency gap as:

$$\Delta = \frac{R_{\text{codec}} - R(D)}{R(D)} \tag{11}$$

The codecs in the WebRTC stack can be ordered by their empirical proximity to $R(D)$ for typical video content at conversational resolutions (720p, 30 fps). Using published Bjøntegaard Delta (BD) rate measurements:

$$R_{\text{H.264}} > R_{\text{VP9}} > R_{\text{AV1}} > R(D) \tag{12}$$

Specifically, the empirical rate savings at equivalent PSNR are approximately:

$$R_{\text{VP9}} \approx 0.65 \cdot R_{\text{H.264}} \quad (\sim 35\% \text{ savings}) \tag{13}$$

$$R_{\text{AV1}} \approx 0.70 \cdot R_{\text{VP9}} \approx 0.46 \cdot R_{\text{H.264}} \quad (\sim 30\% \text{ over VP9}) \tag{14}$$

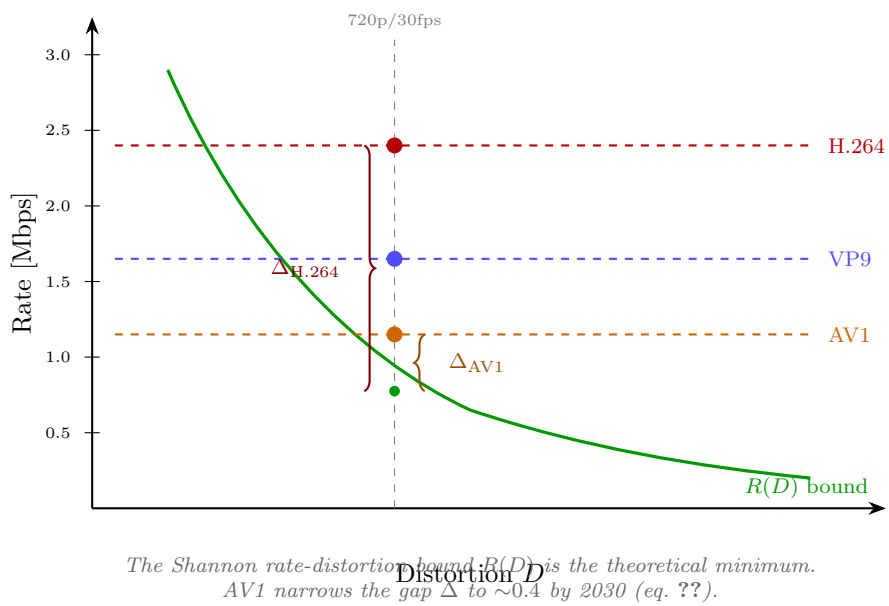


Figure 11: Codec operating rates vs. the Shannon rate-distortion bound $R(D)$ at 720p/30fps. Each codec operates above the theoretical minimum; the gap Δ (eq. **??**) shrinks from H.264 through VP9 to AV1, approaching but never reaching the bound.

[1] The $R(D)$ curve in Figure **??** is the Gaussian memoryless source bound (equation **??**), which is an upper bound on the true $R(D)$ for non-Gaussian sources such as video. Additionally, $R(D)$ assumes zero excess-distortion probability, whereas practical codecs permit a frame-level excess-distortion probability on the order of 10^{-6} to 10^{-8}. Both factors allow codec operating points measured on real content to approach or marginally cross the Gaussian bound.

A.2.1 Codec Entropy and WebCodecs

The WebCodecs API (W3C Working Draft) exposes per-frame codec access, allowing measurement and control of the instantaneous encoding rate. For a video frame f_t at time t, the encoder output has empirical entropy:

$$\hat{H}(f_t) = -\sum_{s \in \mathcal{S}} p(s \mid f_t) \log_2 p(s \mid f_t) \tag{15}$$

where $\mathcal{S}$ is the set of symbols emitted by the entropy coder (CABAC for H.264/AV1, arithmetic coding for VP9). WebCodecs enables applications to observe $|E(f_t)|$ (the encoded frame size in bits) directly, which serves as a practical upper bound on $\hat{H}(f_t)$:

$$\hat{H}(f_t) \leq |E(f_t)| \leq R_{\text{codec}}/\text{fps} \tag{16}$$

A.3 Effective Information Rate by Time Horizon

We now combine transport and source coding measures to estimate the effective information rate for a representative real-time video session: a single 720p/30fps video call with one audio channel.

> **Excursus: Why 720p?** — Throughout this appendix, 720p/30fps is used as a fixed reference for cross-era comparison, not as a prediction that 720p remains the dominant conferencing resolution. The bandwidth surplus quantified below is precisely what enables migration to 1080p (feasible by ∼2028 with AV1 hardware encode at today's 720p bit rates) and 4K (feasible by ∼2030). The 720p baseline makes the efficiency gains legible: the same task gets cheaper, freeing capacity for higher quality or more streams.

Define the *net information rate* as:

$$I_{\text{net}} = R_{\text{codec}} \cdot \eta_{\text{transport}} \cdot (1 - \rho_{\text{overhead}}) \cdot (1 - p_{\text{loss}}) \tag{17}$$

Parameter	**March 2026**	**2027–2028**	**2029–2031**
Median link capacity	120 Mbps	∼150 Mbps	200–300 Mbps
Dominant codec (WebRTC)	VP9 / H.264	VP9 / AV1 (HW)	AV1 (HW default)
Codec rate (720p/30fps)	1.5–2.5 Mbps	1.0–1.8 Mbps	0.7–1.2 Mbps
$\eta_{\text{transport}}$	0.92 (QUIC)	0.93	0.94
ρ_{overhead} (SRTP+QUIC)	0.06	0.05	0.04
Typical loss p	0.01–0.03	0.01–0.02	0.005–0.01
I_{net} (720p video)	1.3–2.2 Mbps	0.9–1.6 Mbps	0.6–1.1 Mbps
I_{net} as % of C_{link}	1.1–1.8%	0.6–1.1%	0.2–0.6%

Table 2: Effective information rate for a 720p/30fps WebRTC video stream. The declining ratio $I_{\text{net}}/C_{\text{link}}$ reflects both improved codec efficiency and growing link capacity, freeing bandwidth for higher resolutions, additional streams, or concurrent data channels.

The declining ratio $I_{\text{net}}/C_{\text{link}}$ in Table **??** is the central quantitative observation: by 2030, a 720p video call consumes less than 0.5% of available link capacity. The surplus capacity enables qualitative changes — 4K video conferencing, multi-stream layouts, or concurrent data channels via WebTransport — that are quantitatively infeasible in 2026.

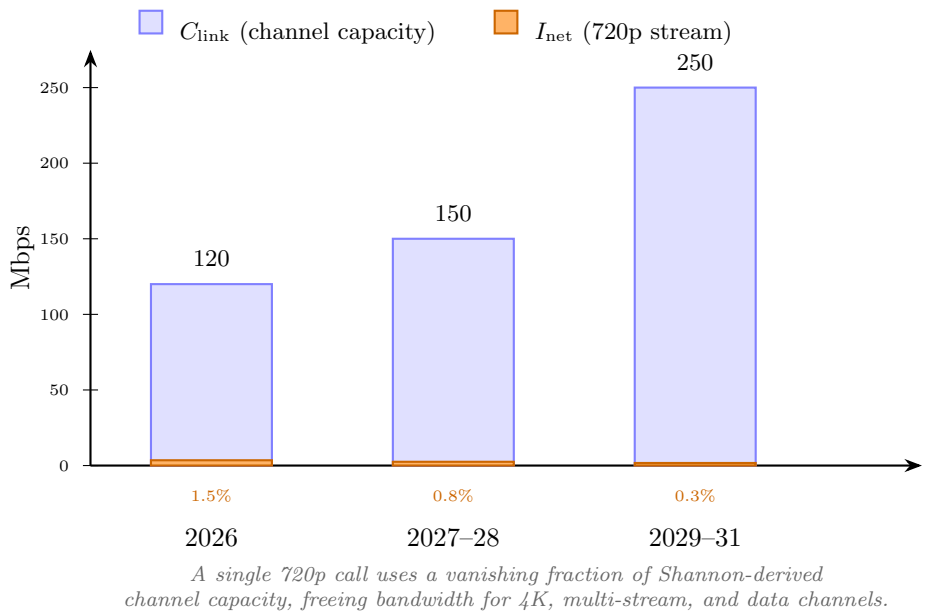


Figure 12: Channel capacity C_{link} vs. effective information rate I_{net} for a single 720p/30fps stream across time horizons. The orange bar (stream rate) is barely visible against the blue bar (capacity), illustrating the growing surplus as codecs improve and bandwidth increases.

A.4 Security Entropy and Key Space

The DTLS-SRTP security architecture (RFCs 8827, 3711) protects all WebRTC media. Information-theoretic security requires that the key entropy exceed the information content of the protected material.

For a DTLS 1.3 handshake with ECDHE on P-256, the key agreement provides:

$$H_{\text{key}} = 128 \text{ bits of security} \tag{18}$$

The SRTP master key is 128 bits (AES-128-GCM), giving an exhaustive search cost of 2^{128} operations. The total information content of a one-hour 720p video call at 2 Mbps is:

$$I_{\text{call}} = 2 \times 10^6 \times 3600 = 7.2 \times 10^9 \text{ bits} \approx 2^{32.7} \text{ bits} \tag{19}$$

The ratio of key entropy to protected content:

$$\frac{H_{\text{key}}}{I_{\text{call}}} = \frac{128}{32.7} \approx 3.91 \tag{20}$$

This ratio remains comfortably above 1 even for much longer sessions. For a continuous 24-hour 4K AV1 stream at 15 Mbps:

$$I_{\text{stream}} = 15 \times 10^6 \times 86400 = 1.296 \times 10^{12} \text{ bits} \approx 2^{40.2} \text{ bits} \tag{21}$$

yielding $H_{\text{key}}/I_{\text{stream}} \approx 3.18$, still well above the threshold.

A.4.1 End-to-End Encryption via Encoded Transforms

The RTCRtpScriptTransform API (Encoded Transforms) enables application-layer E2EE within the WebRTC pipeline. From an information-theoretic perspective, this introduces a second encryption layer with its own key, creating a *nested channel*:

$$X \xrightarrow{K_{\text{E2EE}}} E_1(X) \xrightarrow{K_{\text{SRTP}}} E_2(E_1(X)) \tag{22}$$

The mutual information between the ciphertext and plaintext, given neither key, satisfies:

$$I(X;\, E_2(E_1(X))) \leq \min\bigl(H(K_{\text{E2EE}})^{-1},\ H(K_{\text{SRTP}})^{-1}\bigr) \cdot I_{\text{call}} \tag{23}$$

In practice, with $H(K_{\text{E2EE}}) = H(K_{\text{SRTP}}) = 128$ bits, this quantity is negligible ($\ll 1$ bit for any practical session length), confirming that the nested encryption provides information-theoretic confidentiality well beyond computational security guarantees.

Today (March 2026), Encoded Transforms remain Chrome-only. By 2027–2028 (Section 8.3), cross-browser standardization will make this nested channel universally available. By 2029–2031, SFrame for group calls will extend this to n-party sessions, where the mutual information between any non-member and the group plaintext remains negligible:

$$I(X;\, E(X) \mid K_i \notin \{K_1, \ldots, K_n\}) \approx 0 \tag{24}$$

A.5 WebTransport: Unreliable Datagrams and Channel Capacity

WebTransport (RFC 9297) introduces unreliable datagrams to browser-server communication. For latency-sensitive applications (game state, sensor telemetry), the relevant measure is not throughput but *freshness-weighted information rate*. Define the value of a datagram as exponentially decaying with delivery latency δ:

$$V(\delta) = e^{-\lambda\delta} \tag{25}$$

where $\lambda > 0$ is an application-specific decay rate. The effective information rate under unreliable delivery is:

$$I_{\text{WT}} = R_{\text{send}} \cdot (1 - p_{\text{loss}}) \cdot \mathbb{E}\Bigl[e^{-\lambda\delta}\Bigr] \tag{26}$$

Under reliable delivery (TCP/WebSocket), a retransmission delays subsequent data, so:

$$I_{\text{WS}} = R_{\text{send}} \cdot \mathbb{E}\Bigl[e^{-\lambda(\delta + p\cdot 2\tau)}\Bigr] \tag{27}$$

The ratio $I_{\text{WT}}/I_{\text{WS}}$ exceeds 1 when:

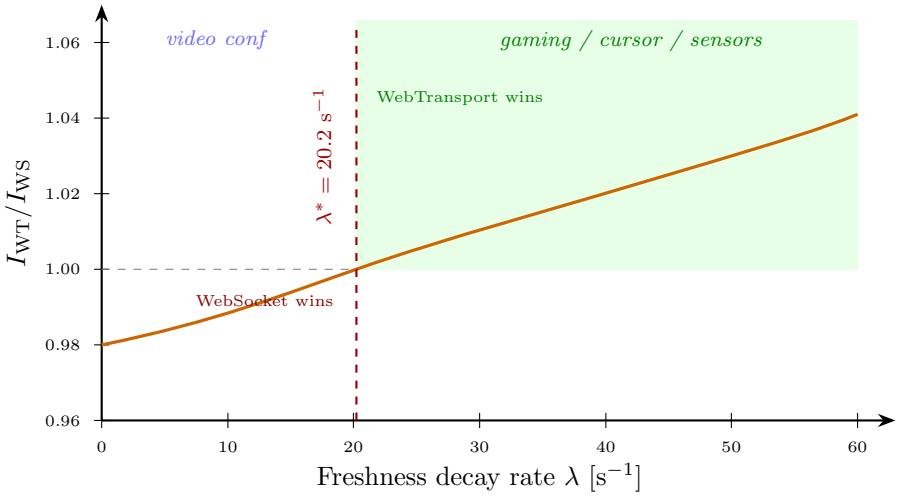
$$(1-p) \cdot e^{\lambda \cdot p \cdot 2\tau} > 1 \quad \Longleftrightarrow \quad \lambda > \frac{-\ln(1-p)}{2p\tau} \tag{28}$$

For $p = 0.02$ and $\tau = 25$ ms, this gives $\lambda > 20.2\ \text{s}^{-1}$ (half-life < 34 ms).

Excursus: Why $p = 0.02$ and $\tau = 25$ ms? — These values represent median fixed-broadband conditions in 2026 (Table **??**). On mobile or congested links ($p \approx 0.05$, $\tau \approx 50$ ms), the threshold λ^* drops to $\sim 10\,\text{s}^{-1}$, broadening WebTransport's advantage to applications with data half-lives up to ~ 70 ms. The fixed-broadband reference is thus conservative: mobile scenarios favor unreliable delivery even more strongly.

Any application where data older than ∼30 ms is useless benefits from WebTransport's unreliable mode over WebSocket — precisely the regime of interactive gaming, live cursor tracking, and real-time sensor feeds.

As of March 2026, Safari does not support WebTransport (Section 5), limiting coverage to ∼82%. By 2027–2028 (Section 8.2), universal support will allow applications to rely on (**??**) without fallback paths.



*For $p = 0.02$, $\tau = 25$ ms. Applications where data older than ∼30 ms is stale gain from unreliable delivery (eq. **??**).*

Figure 13: Freshness-weighted information rate ratio $I_{\mathrm{WT}}/I_{\mathrm{WS}}$ as a function of decay rate λ. Above the threshold λ^*, WebTransport's unreliable datagrams deliver more useful information per unit time than WebSocket's reliable delivery, because retransmission wastes capacity on stale data.

A.6 Projections Summary and Practical Implications

Measure	**March 2026**	**2027–2028**	**2029–2031**
Channel capacity C_{link}	120 Mbps	150 Mbps	250 Mbps
HOL blocking factor $(1-p)^{n-2}$ at $n=8$	0.94 ($p=0.01$)	0.94	0.97 ($p=0.005$)
Source coding gap Δ	VP9: ∼0.8	AV1 HW: ∼0.5	AV1 opt: ∼0.4
I_{net} per 720p stream	1.3–2.2 Mbps	0.9–1.6 Mbps	0.6–1.1 Mbps
Max 720p streams in C_{link}	∼55–90	∼95–165	∼230–415
E2EE availability	Chrome only	Cross-browser	SFrame n-party
WebTransport coverage	82%	∼100%	100%
Freshness threshold λ^*	20.2 s^{-1}	∼18 s^{-1}	∼15 s^{-1}

Table 3: Information-theoretic measures across time horizons. Δ is the coding efficiency gap (**??**), I_{net} is the net information rate (**??**), and λ^* is the freshness decay threshold above which WebTransport dominates WebSocket (**??**).

The trajectory is clear: the combination of improving source coding (AV1 hardware), expanding channel capacity (broadband growth), and maturing transport protocols (QUIC universal,

WebTransport universal) drives the *information cost per unit of communicated experience* steadily downward. The remaining question is not whether the Shannon capacity suffices — it already does, by orders of magnitude — but how efficiently the protocol stack converts that capacity into delivered, fresh, secure information at the application layer. By 2030, the gap between theoretical capacity and realized throughput will be the narrowest it has ever been for browser-based real-time communication.

A.6.1 What the Theory Means for Applications

The equations and figures in this appendix are not purely academic. Each theoretical result maps to a concrete design decision that application developers face today and over the next five years.

Codec selection determines bandwidth cost, not quality. The rate-distortion analysis (Section A.2, Figure **??**) shows that AV1 operates at roughly 46% of H.264's bit rate for equivalent perceptual quality at 720p. In practice, this means a WebRTC application switching from H.264 to AV1 can either halve its bandwidth consumption at the same quality or substantially increase resolution (e.g., 720p to 1080p) within the same bandwidth envelope. By 2030, with AV1 hardware encoding ubiquitous and the coding gap Δ approaching 0.4, further codec improvements yield diminishing returns — the stack will be operating close enough to the Shannon rate-distortion bound that the next generation of efficiency gains will come from transport and architecture, not codecs alone.

QUIC's advantage is situational but decisive on bad networks. The head-of-line blocking analysis (Section A.1.1, Figure **??**) shows that QUIC's per-stream throughput advantage over TCP is modest on good networks (6% at 1% loss) but substantial on degraded ones (34% at 5% loss). The practical implication is that applications targeting mobile users, emerging markets, or congested enterprise networks — precisely the environments where real-time communication is hardest — benefit disproportionately from QUIC-based transports. For a video conference with 8 participants on a cellular connection experiencing 3–5% packet loss, the difference between QUIC and TCP is the difference between usable and degraded video.

Connection latency is a solved problem for returning users. The handshake analysis (Section A.1.2) shows that QUIC 0-RTT eliminates connection establishment overhead entirely for resumed connections. For a WebTransport application that a user opens daily — a collaboration tool, a live dashboard, a recurring meeting client — the time to first useful byte drops to zero. The 1.125 MB of wasted capacity during a TCP+TLS handshake on a 120 Mbps link may seem small, but for latency-sensitive applications where the first 100 ms of perceived responsiveness determines user experience, the difference is measurable.

Bandwidth surplus enables architectural ambition. The effective information rate analysis (Section A.3, Figure **??**) reveals the most consequential practical finding: a single 720p video stream consumes a vanishing fraction of available link capacity (under 0.5% by 2030). This surplus is not merely "headroom" — it is the resource that makes new application architectures feasible. A 250 Mbps link can theoretically carry 230–415 simultaneous 720p streams. In practice, this means applications can afford to run multiple concurrent media flows (video, screen share, spatial audio), data channels (collaborative editing, file transfer), and telemetry streams without contending for bandwidth. The constraint shifts from "can the network carry this?" to "can the application manage this complexity?"

End-to-end encryption adds negligible overhead. The security analysis (Section A.4) con-

firms that the nested DTLS-SRTP plus E2EE architecture imposes no meaningful information-theoretic cost. The key entropy (2^{128}) exceeds the information content of any practical session by a factor of at least 3. For application developers, this means E2EE can be treated as a default rather than a premium feature — there is no bandwidth, latency, or capacity penalty for encrypting at the application layer on top of the transport-layer encryption that WebRTC already mandates. The only constraint is API availability, which moves from Chrome-only (2026) to universal (2028).

WebTransport unlocks real-time data that WebSocket cannot serve. The freshness analysis (Section A.5, Figure **??**) quantifies when unreliable delivery outperforms reliable delivery: any data with a useful half-life below approximately 34 ms benefits from WebTransport's unreliable datagrams. This covers a wide class of interactive applications — game state synchronization, live cursor and pointer sharing, real-time sensor feeds, collaborative whiteboard strokes — where retransmitting a stale update is worse than dropping it. Before WebTransport, these applications had to choose between WebSocket (reliable but stale under loss) and WebRTC DataChannel (unreliable mode available but requiring the full peer-to-peer ICE/DTLS machinery). WebTransport provides the unreliable mode with a simple client-server model over HTTP/3, removing the architectural overhead.

The overall picture: the stack converges on efficiency. Taken together, the information-theoretic measures trace a consistent trajectory. The protocol stack is converging toward the theoretical limits established by Shannon: codecs approach the rate-distortion bound, transports eliminate unnecessary blocking and handshake overhead, and encryption adds negligible cost. By 2030, the practical distance between what the standards deliver and what information theory permits will be the smallest it has ever been for browser-based real-time communication. The consequence for application developers is that the infrastructure — standards, hardware, and networks — will no longer be the binding constraint. The remaining challenge is purely architectural: how to compose these efficient, universal primitives into applications that fully exploit the capacity and low latency they provide.

B Video Compression Fundamentals

Appendix **??** quantified the coding efficiency gap Δ (eq. **??**) for each codec generation, treating codecs as black boxes characterized by their distance from the Shannon rate-distortion bound. This appendix opens those black boxes. We examine the signal processing and coding techniques that determine Δ: how video codecs exploit spatial, temporal, and perceptual redundancy, what distinguishes one codec generation from the next, and why computational complexity—not algorithmic capability—remains the binding constraint for real-time web communication.

B.1 Why Video Is Compressible

A raw digital video signal carries far more data than information. Three fundamental redundancies make compression ratios of 200:1 to 500:1 achievable at perceptually acceptable quality.

B.1.1 Spatial Redundancy

Natural images exhibit strong local correlation. The autocorrelation function for a typical image row can be modeled as a first-order Markov process:

$$R_{xx}(k) = \sigma^2 \rho^{|k|} \tag{29}$$

where σ^2 is the pixel variance and $\rho \in [0.90, 0.97]$ for natural content. The separable two-dimensional extension is:

$$R_{xx}(k, l) = \sigma^2 \, \rho_h^{|k|} \, \rho_v^{|l|} \tag{30}$$

The power spectral density—the Fourier transform of (**??**)—is concentrated at low spatial frequencies, meaning most of the signal energy occupies a small fraction of the frequency domain. This concentration is the foundation of transform coding (Section **??**).

B.1.2 Temporal Redundancy

Consecutive video frames are highly correlated. For typical conversational video at 30 fps, the inter-frame correlation coefficient is:

$$\mathrm{Corr}(f_t, f_{t-1}) \approx 0.90\text{–}0.98 \tag{31}$$

Most pixels change little between frames; only regions containing motion, lighting changes, or scene cuts carry significant new information. Motion estimation (Section **??**) exploits this by predicting each block from a displaced version of a reference frame, transmitting only the prediction residual.

B.1.3 Perceptual Irrelevance

The human visual system (HVS) has lower spatial acuity for chrominance than luminance. This asymmetry motivates *chroma subsampling*: representing color difference channels at reduced resolution relative to the luma channel.

> **Excursus: Why 4:2:0?** — All WebRTC codecs operate on YCbCr 4:2:0 video, where each chroma channel is sampled at half the horizontal and half the vertical resolution of luma. This reduces the sample count by a factor of $1 + 0.25 + 0.25 = 1.5$ relative to $1 + 1 + 1 = 3$ for full-resolution 4:4:4 color, a $2\times$ saving before any coding begins. Perceptual experiments confirm that 4:2:0 is essentially indistinguishable from 4:4:4 at typical viewing distances for video conferencing—the HVS cannot resolve the missing chroma detail.

B.1.4 The Compression Imperative

A raw 720p/30fps 4:2:0 video stream at 8 bits per sample requires:

$$R_{\text{raw}} = 1280 \times 720 \times 1.5 \times 8 \times 30 = 331.8 \text{ Mbps} \tag{32}$$

This exceeds the 120 Mbps median link capacity cited in Table **??**. The Shannon rate-distortion bound (eq. **??**) yields approximately 0.7–1.5 Mbps at perceptually acceptable distortion—a compression ratio of 220:1 to 475:1. Video compression is not optional for real-time communication; it is a prerequisite.

The first-order entropy of raw pixel values $H(X)$ greatly exceeds the conditional entropy given neighboring samples:

$$H(X) \gg H(X \mid X_{\text{neighbors}}) \tag{33}$$

This gap—the mutual information between a pixel and its neighbors—represents the exploitable redundancy. The entire hybrid coding framework (Section **??**) is designed to extract this mutual information through prediction, leaving only the irreducible residual for transform coding and quantization.

B.2 The Hybrid Coding Framework

Every major video codec since H.261 (1988)—including every codec in the WebRTC stack—is a variation on the *block-based hybrid coding* framework. The encoder partitions each frame into blocks, predicts each block from previously coded data, transforms and quantizes the prediction residual, and entropy codes the result.

B.2.1 Encoder Signal Flow

For a block at position (x, y) in frame f_t, the encoder computes a prediction $\hat{f}_{\text{pred}}(x, y)$ from either the current frame (intra prediction) or a reference frame (inter prediction). The prediction residual is:

$$e(x, y) = f_t(x, y) - \hat{f}_{\text{pred}}(x, y) \tag{34}$$

The residual is transformed, quantized, and entropy coded to produce the bitstream. The decoder reconstructs the frame as:

$$\hat{f}(x, y) = \hat{e}(x, y) + \hat{f}_{\text{pred}}(x, y) \tag{35}$$

where $\hat{e}$ is the inverse-transformed, dequantized residual.

B.2.2 Rate-Distortion Optimization

Every coding decision—block partition, prediction mode, motion vector, transform type, quantization level—is made by minimizing a Lagrangian rate-distortion cost:

$$J(m) = D(m) + \lambda \cdot R(m) \tag{36}$$

where $m \in \mathcal{M}$ is a coding mode, $D(m)$ is the distortion (typically sum of squared errors), $R(m)$ is the bit cost, and λ is a Lagrange multiplier controlling the rate-distortion trade-off. The optimal mode satisfies:

$$m^* = \arg\min_{m \in \mathcal{M}} [D(m) + \lambda \cdot R(m)] \tag{37}$$

The Lagrange multiplier relates to the quantization parameter (QP) via an empirically validated exponential relationship:

$$\lambda = c \cdot 2^{(QP-12)/3} \tag{38}$$

where c is a codec-specific constant. This relationship ensures that finer quantization (lower QP) assigns higher cost to bits, favoring modes that reduce distortion, while coarser quantization (higher QP) favors modes that reduce rate.

The four sub-optimal components of Figure **??**—prediction, transform, quantization, and entropy coding—each contribute to Δ:

$$\Delta \approx \Delta_{\text{pred}} + \Delta_{\text{transform}} + \Delta_{\text{quant}} + \Delta_{\text{entropy}} \tag{39}$$

Generational codec improvements reduce one or more of these terms. The codec lineages in Sections **??**–**??** trace exactly which tools target which terms.

B.3 Transforms

The transform stage decorrelates the prediction residual, concentrating energy into a few coefficients that can be efficiently quantized and coded.

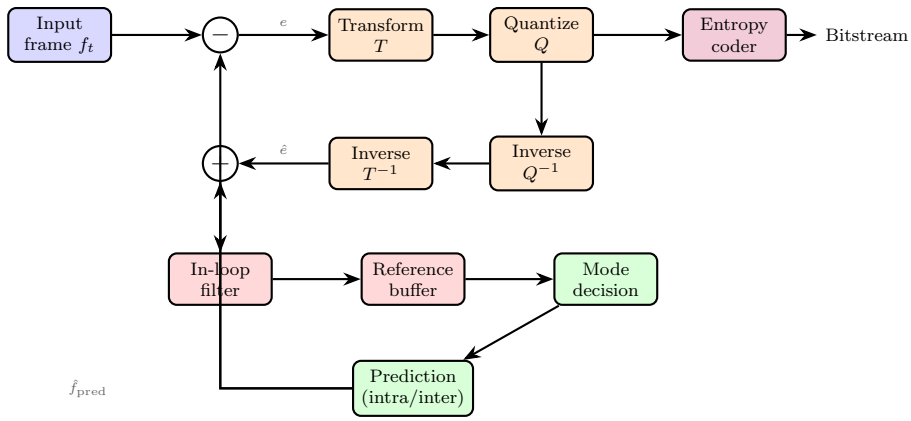


Figure 14: Block-based hybrid encoder architecture. Every codec from H.264 through VVC follows this structure: predict each block from previously coded data (green), transform and quantize the residual (orange), entropy code the result (purple), then reconstruct and filter for future reference (red). The coding efficiency gap Δ (eq. **??**) arises from sub-optimal prediction, imperfect energy compaction, quantization beyond the rate-distortion bound, and entropy coding overhead.

B.3.1 The KLT and Its DCT Approximation

The optimal decorrelating transform for a stationary source is the *Karhunen-Loève Transform* (KLT), whose basis vectors are the eigenvectors of the input autocorrelation matrix $\mathbf{R}_{xx}$. For a first-order Markov source with correlation ρ (eq. **??**), the KLT basis converges to the Type-II DCT as block size grows:

$$C_k(n) = \alpha_k \cos\left[\frac{\pi k(2n+1)}{2N}\right], \quad k, n = 0, \ldots, N-1 \tag{40}$$

where $\alpha_0 = \sqrt{1/N}$ and $\alpha_k = \sqrt{2/N}$ for $k > 0$.

The energy compaction efficiency—the fraction of total energy captured by the first K of N coefficients—is:

$$\eta(K, N) = \frac{\sum_{k=0}^{K-1} \sigma_k^2}{\sum_{k=0}^{N-1} \sigma_k^2} \tag{41}$$

where σ_k^2 is the variance of the k-th transform coefficient. For a typical intra residual with $\rho = 0.95$ and $N = 8$, the first 3 DCT coefficients capture approximately 95% of the energy. This concentration enables aggressive quantization of high-frequency coefficients with minimal perceptual impact.

B.3.2 Integer Transforms in Practice

Practical codecs use integer approximations of the DCT to ensure bit-exact encoder-decoder agreement. H.264 introduced a 4×4 integer transform:

$$\mathbf{H}_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -1 & -2 \\ 1 & -1 & -1 & 1 \\ 1 & -2 & 2 & -1 \end{pmatrix} \tag{42}$$

The non-unity scaling factors are absorbed into the quantization step, preserving exact integer arithmetic throughout the encode-decode loop.

Each codec generation expands the transform toolkit:

Codec	Transform sizes	Transform types	Selection method
H.264	4×4, 8×8	Integer DCT	Fixed per block size
VP9	4×4 to 32×32	DCT, ADST	Row/column adaptive
HEVC	4×4 to 32×32	Integer DCT, DST (4×4 intra)	Mode-dependent
AV1	4×4 to 64×64	DCT, ADST, FlipADST, Identity	RD-optimized per block
VVC	4×4 to 64×64	DCT-II, DST-VII + LFNST	RD-optimized + secondary

Table 4: Transform sizes and types across codec generations. AV1 applies four 1D transform types independently to rows and columns, yielding 16 possible 2D combinations per block. VVC adds a non-separable secondary transform (LFNST) applied to low-frequency coefficients after the primary separable transform.

The Asymmetric Discrete Sine Transform (ADST) used in VP9 and AV1 is particularly effective for intra-predicted residuals, which tend to have larger magnitudes near the prediction boundary. AV1's FlipADST extends this to blocks where the large residuals are on the opposite edge, while the Identity transform (skip) handles blocks where the residual is already well-distributed in the spatial domain.

The progression from fixed transforms (H.264) to RD-optimized multi-type selection (AV1, VVC) directly reduces $\Delta_{\text{transform}}$ in (**??**): better energy compaction means fewer bits are needed to represent the same residual at a given distortion level.

B.4 Motion Estimation and Inter Prediction

Inter prediction exploits temporal redundancy by predicting blocks in the current frame from previously coded reference frames. This is the single largest source of compression gain—the prediction residual (eq. **??**) typically has 10–20× less energy than the original signal.

B.4.1 Block Matching and Motion Vectors

The encoder searches for the best-matching region in a reference frame by minimizing a cost function over candidate displacement vectors $\mathbf{d} = (d_x, d_y)$:

$$\text{SAD}(\mathbf{d}) = \sum_{(x,y)\in B} |f_t(x, y) - f_{t-1}(x + d_x,\, y + d_y)| \tag{43}$$

The Sum of Absolute Differences (SAD) is the simplest distortion measure. A better rate-distortion proxy is the Sum of Absolute Transformed Differences (SATD), which applies the Hadamard transform before summation:

$$\text{SATD}(\mathbf{d}) = \sum |\mathbf{H} \cdot (f_t - f_{t-1}(\mathbf{d}))| \tag{44}$$

SATD approximates the post-transform energy and thus better predicts the actual coded bit cost. Rate-constrained motion estimation augments the distortion with a motion vector coding cost:

$$\mathbf{d}^* = \arg\min_{\mathbf{d}} \left[\mathrm{SATD}(\mathbf{d}) + \lambda_{\mathrm{motion}} \cdot R(\mathbf{d})\right] \tag{45}$$

where $R(\mathbf{d})$ is the bit cost of encoding the motion vector relative to a predictor.

B.4.2 Sub-Pixel Interpolation

Motion in natural video rarely aligns to integer pixel positions. All modern codecs perform sub-pixel motion compensation by interpolating reference frame samples at fractional positions.

H.264 uses a 6-tap filter for half-pixel luma interpolation:

$$h_{\frac{1}{2}} = \{1, -5, 20, 20, -5, 1\}/32 \tag{46}$$

with quarter-pixel positions obtained by bilinear averaging of integer and half-pixel samples. HEVC extends this to an 8-tap filter for half-pixel and a 7-tap filter for quarter-pixel positions. AV1 introduces a switchable filter bank with three 8-tap options—Regular (sharp), Smooth (low-pass), and Sharp (high-frequency preserving)—selected per block via RD optimization. VVC adds DMVR (Decoder-side Motion Vector Refinement) and BDOF (Bi-Directional Optical Flow) for sub-pixel refinement at the decoder without additional signaling bits.

The progression from H.264's fixed 6-tap filter to AV1's switchable bank reduces Δ_{pred}: better interpolation means smaller prediction residuals, which require fewer bits.

B.4.3 Advanced Inter Prediction Tools

Beyond basic block matching, each codec generation adds tools to handle complex motion and occlusion:

- **Multiple reference frames.** H.264 introduced up to 16 reference frames; practical WebRTC encoders use 1–3 for latency reasons. More references improve prediction for periodic motion and uncovered backgrounds.
- **Bi-prediction.** A weighted average of predictions from two reference frames, reducing noise and handling gradual scene changes. Available from H.264 onward (in B-frames) and in AV1's compound modes.
- **Compound prediction (AV1).** Wedge masks and difference-weighted blending combine two predictions at the block level, handling occlusion boundaries where a single motion model fails.
- **Overlapped Block Motion Compensation (AV1).** Blends predictions from neighboring blocks at motion boundaries, reducing blocking artifacts without post-processing.
- **Warped motion (AV1).** An affine model with up to 6 parameters handles rotation, zoom, and perspective—common in screen sharing and camera motion.
- **VVC additions.** Geometric Partitioning Mode (non-rectangular block splits for motion boundaries), BDOF (optical flow refinement at the decoder), and DMVR (decoder-side MV refinement using bilateral matching).

Excursus: Motion estimation in real-time encoding. — Full-search block matching has complexity $O(N^2 \cdot S^2)$ where N is the block dimension and S is the search range in pixels. For a 64×64 block with $S = 64$, this is ∼17 million SAD operations per block—far too expensive for real-time encoding. WebRTC encoders use hierarchical search patterns (diamond, hexagon) that reduce complexity to $O(N^2 \cdot \log S)$ with typically $< 5\%$ rate-distortion penalty relative to full search. This is why real-time encoders use only a subset of each codec's inter prediction tools: the full tool set is available for offline encoding, but the real-time constraint forces aggressive pruning of the mode decision search space.

B.5 Quantization

Quantization is the sole lossy step in the hybrid coding pipeline. It maps continuous-valued (or high-precision integer) transform coefficients to a discrete set of reconstruction levels, introducing controlled information loss to achieve the target bit rate.

B.5.1 Scalar Quantization

Uniform scalar quantization maps coefficient c to a quantization index:

$$q = \text{round}\left(\frac{c}{Q_{\text{step}}}\right), \quad \hat{c} = q \cdot Q_{\text{step}} \tag{47}$$

Modern codecs use *dead-zone quantization*, where coefficients near zero are mapped to zero more aggressively than uniform rounding predicts:

$$q = \text{sign}(c) \cdot \left\lfloor \frac{|c| - f \cdot Q_{\text{step}}}{Q_{\text{step}}} \right\rfloor \tag{48}$$

where $f \in [0, 0.5)$ controls the dead-zone width. Dead-zone quantization increases sparsity—the fraction of zero-valued coefficients—which dramatically improves entropy coding efficiency.

B.5.2 QP-to-Step-Size Mapping

The quantization parameter (QP) controls the step size via an exponential relationship. In the H.264/HEVC/VVC family:

$$Q_{\text{step}} = Q_0 \cdot 2^{(QP - QP_0)/6} \tag{49}$$

A 6-unit increase in QP doubles the step size, approximately doubling the bit rate. AV1 uses a similar but codec-specific mapping with separate QP values for DC and AC coefficients, plus per-segment QP adjustment for region-of-interest coding.

B.5.3 Quantization Distortion

For a single coefficient under uniform quantization at high rate, the distortion is:

$$D_q = \frac{Q_{\text{step}}^2}{12} \tag{50}$$

For a block of N transform coefficients where K survive quantization (nonzero) and $N - K$ are quantized to zero:

$$D_{\text{block}} \approx K \cdot \frac{Q_{\text{step}}^2}{12} + \sum_{k \notin K} c_k^2 \tag{51}$$

The second term represents the energy of coefficients eliminated by the dead zone—information permanently discarded. The QP is thus the primary control knob for navigating the rate-distortion curve in Figure **??**. The WebCodecs `VideoEncoder` API exposes this control via the `quantizer` configuration parameter, giving JavaScript applications direct access to this fundamental trade-off.

B.6 Entropy Coding

After quantization, the surviving coefficients and all side information (motion vectors, prediction modes, block partitions) must be losslessly compressed. The entropy coder converts these symbols into a bitstream, approaching the conditional entropy $H(X|\text{context})$ as closely as possible. The gap between the entropy coder's output rate and the true conditional entropy is Δ_{entropy} in (**??**).

B.6.1 Context-Adaptive Binary Arithmetic Coding (CABAC)

CABAC, used in H.264, HEVC, and VVC, encodes each syntax element as a sequence of binary decisions (bins). For each bin, the arithmetic coder subdivides the current interval $[L, H)$ based on the estimated probability p:

$$[L, H) \to \begin{cases} [L,\ L + p \cdot (H - L)) & \text{if bin} = 0 \\ [L + p \cdot (H - L),\ H) & \text{if bin} = 1 \end{cases} \tag{52}$$

The theoretical output length approaches the entropy:

$$\ell \to \sum_{i=1}^{n} -\log_2 p(b_i \mid \text{ctx}_i) \tag{53}$$

CABAC achieves redundancy below 0.01 bits/symbol relative to the true conditional entropy—near-optimal compression.

Context modeling is the key to CABAC's effectiveness. Each bin type has an associated context, whose probability is updated after each observation via a state-machine-based exponential moving average:

$$p_{k+1} = p_k + \alpha(s_k) \cdot (b_k - p_k) \tag{54}$$

where $\alpha(s_k)$ is a state-dependent adaptation rate, indexed by a lookup table with 64 states (H.264) or 128 states (HEVC/VVC). This design avoids floating-point arithmetic entirely.

CABAC also provides a *bypass mode* for bins with approximately uniform probability (e.g., sign bits, high-magnitude coefficient levels), which skips the context lookup and probability estimation for throughput.

B.6.2 Multi-Symbol Arithmetic Coding in AV1

AV1 departs from binary arithmetic coding. Its entropy coder operates on multi-symbol alphabets using CDF (Cumulative Distribution Function) tables:

$$\text{Encode symbol } s: \quad [L, H) \to \left[L + \frac{\text{CDF}(s-1)}{M} \cdot (H - L),\ L + \frac{\text{CDF}(s)}{M} \cdot (H - L)\right) \tag{55}$$

where M is the total frequency count. CDF tables are adapted after each symbol:

$$\text{CDF}_{k+1}(s) = \text{CDF}_k(s) + \left\lfloor \frac{\text{target}(s) - \text{CDF}_k(s)}{n} \right\rfloor \tag{56}$$

where n controls the adaptation rate. Multi-symbol coding is more efficient than CABAC's binarization for syntax elements with moderate alphabet sizes (e.g., intra prediction modes, transform types), as it avoids the overhead of converting each symbol into a binary tree of decisions.

Feature	**H.264**	**HEVC**	**VP9**	**AV1**	**VVC**
Method	Binary AC	Binary AC	Boolean AC	Multi-sym AC	Binary AC
Contexts	∼400	∼200	∼900	∼300	∼700
Adaptation	64-state LUT	128-state LUT	Counter	CDF update	128-state LUT
Bypass mode	Yes	Yes	N/A	N/A	Yes
Parallelism	Slice	WPP / Tiles	Tiles	Tiles	WPP / Tiles

Table 5: Entropy coding comparison across codecs. HEVC reduced context count relative to H.264 for throughput; VVC increased it again for improved compression. AV1's multi-symbol approach operates on full alphabets rather than binary decompositions.

The frame entropy $\hat{H}(f_t)$ from eq. (**??**) is ultimately determined by how well the entropy coder's probability model matches the true symbol distribution. Both CABAC and AV1's multi-symbol coder achieve near-zero entropy coding gaps; the remaining Δ_{entropy} arises primarily from context model mismatch on non-stationary content (scene changes, sudden motion onset).

B.7 In-Loop Filtering

Block-based coding introduces visible artifacts at block boundaries: discontinuities from independent quantization of adjacent blocks, and ringing from coefficient truncation. In-loop filters reduce these artifacts *before* the reconstructed frame enters the reference buffer, improving prediction quality for subsequent frames and thus reducing Δ_{pred}.

- **Deblocking filter (all codecs).** Smooths discontinuities at block edges, with filter strength adapted to the quantization level and local gradient. Introduced in H.264; refined in each subsequent codec.
- **Sample Adaptive Offset—SAO (HEVC).** Classifies reconstructed samples by intensity or edge direction and applies per-class offsets to reduce systematic bias introduced by quantization.
- **Constrained Directional Enhancement Filter—CDEF (AV1).** A directional filter that identifies edge orientation per 8×8 block and applies filtering only along the edge, preserving sharpness while reducing ringing.
- **Loop Restoration Filter (AV1).** A switchable per-tile filter choosing between a Wiener filter (FIR designed to minimize MSE between original and reconstruction) and a Self-Guided filter (non-linear denoising based on guided image filtering). This is AV1's most powerful quality enhancement tool.
- **Adaptive Loop Filter—ALF (VVC).** A diamond-shaped filter with coefficients signaled per frame, applied adaptively per 4×4 block based on local classification. Subsumes HEVC's SAO functionality.

The evolution from simple deblocking (H.264) to multi-stage adaptive filtering (AV1: deblocking

→ CDEF → loop restoration) exemplifies how each codec generation reduces Δ_{pred} by improving reference frame quality.

B.8 Codec Lineage: MPEG/ITU Family

The MPEG/ITU standardization track has produced three codec generations relevant to the 2021–2031 timeframe of this document.

B.8.1 H.264/AVC (2003)

H.264 defined the modern era of video compression and remains the most widely deployed codec. Its key innovations over its predecessors (H.263, MPEG-2):

- Variable block sizes: 16×16 macroblock partitioned into 16×8, 8×16, 8×8, 8×4, 4×8, and 4×4 sub-partitions for motion compensation.
- Quarter-pixel motion compensation with 6-tap interpolation (eq. **??**).
- Context-adaptive intra prediction: 9 angular modes for 4×4 blocks, 4 modes for 16×16.
- Dual entropy coding: CAVLC (simpler, faster) and CABAC (Section **??**; 10–15% better compression).
- In-loop deblocking filter, mandatory for the decoder.
- Multiple reference frames (up to 16) and weighted prediction.

Three profiles are relevant to WebRTC: **Constrained Baseline** (no CABAC, no B-frames, lowest complexity—the mandatory WebRTC profile), **Main** (CABAC, B-frames, ∼10% better compression), and **High** (8×8 transform, adaptive quantization matrices, ∼15% better).

At the 720p/30fps reference point, H.264 Constrained Baseline operates at approximately 2.0–2.5 Mbps for "good" quality (PSNR ∼37 dB), corresponding to $\Delta_{\text{H.264}} \approx 2.1$ from Appendix **??**. H.264 was the only codec Safari supported in 2021 (Section 3.1), and its higher Δ is directly responsible for the bandwidth disparity between Safari-only and VP9-capable sessions documented in the main text.

B.8.2 H.265/HEVC (2013)

HEVC replaced H.264's fixed 16×16 macroblock with a recursive Coding Tree Unit (CTU) of up to 64×64 pixels, using quad-tree partitioning down to 8×8. Key advances:

- 35 intra prediction angular directions (vs. H.264's 9).
- Transform sizes from 4×4 to 32×32, with DST for 4×4 intra blocks.
- Advanced Motion Vector Prediction (AMVP): two MVP candidates selected from spatial and temporal neighbors, replacing H.264's less structured motion vector prediction.
- Merge mode: inherits the full motion information (MV, reference index, prediction direction) from a neighboring block, requiring zero additional motion bits.
- Sample Adaptive Offset (SAO) in-loop filter.
- Parallel processing: Wavefront Parallel Processing (WPP) and tiles.

HEVC achieves approximately 40% rate reduction over H.264 at equivalent PSNR. However, HEVC is absent from WebRTC entirely—not due to technical limitations, but due to patent licensing.

> **Excursus: Why no HEVC in WebRTC?** — HEVC's patent licensing is fragmented across three separate pools (MPEG-LA, HEVC Advance, Velos Media) plus independent licensors who are not members of any pool. The resulting uncertainty about total royalty cost—estimated at \$0.20–\$0.40 per device for some use cases—led Google, Mozilla, Cisco, and others to form the Alliance for Open Media (AOM) in 2015 and develop AV1 as a royalty-free alternative. This patent-driven fork is why the WebRTC ecosystem followed the VP8→VP9→AV1 path rather than H.264→HEVC→VVC, and why the two codec families must be understood in parallel.

B.8.3 H.266/VVC (2020)

VVC extends HEVC's quad-tree with binary and ternary splits, creating a *multi-type tree* (MTT) that enables asymmetric block shapes (e.g., 32×8, 8×32, 32×4). Additional innovations:

- 67 intra prediction directions plus Matrix-based Intra Prediction (MIP), which applies a learned linear transform to neighboring samples.
- Multiple Transform Selection (MTS): explicit signaling of DCT-II or DST-VII per block, plus the Low-Frequency Non-Separable Transform (LFNST)—a 16×16 or 8×8 non-separable transform applied to low-frequency coefficients after the primary separable transform.
- Affine motion model: 4-parameter (translation + rotation/scaling) or 6-parameter (full affine) motion, derived from control point MVs at block corners.
- Decoder-side tools: DMVR refines motion vectors at the decoder using bilateral template matching; BDOF applies optical flow equations to refine bi-predicted samples, both without additional signaling bits.
- Geometric Partitioning Mode: divides a block diagonally for separate motion compensation of each partition, targeting motion boundaries.
- Adaptive Loop Filter (ALF): a diamond-shaped Wiener filter with coefficients signaled per frame and applied adaptively per 4×4 block.

VVC achieves approximately 30–50% rate reduction over HEVC, placing it at $\Delta_{\mathrm{VVC}} \approx 0.3$. Its relevance to this document is forward-looking: VVC may appear in WebCodecs codec registries by ∼2030 if licensing concerns are resolved, and its tool set represents the current efficiency frontier against which AV1's performance should be measured.

B.9 Codec Lineage: Google/AOM Family

The royalty-free codec family dominates the WebRTC ecosystem. Its development was driven by the patent licensing concerns described in the HEVC excursus above.

B.9.1 VP8 (2010)

VP8 was Google's first open-source video codec, released alongside the WebM container. Its design is conservative:

- 16×16 macroblocks with 4×4 subblock partitions.

- 4×4 Walsh-Hadamard Transform (WHT) for DC coefficients, 4×4 DCT for AC coefficients.
- Boolean arithmetic coder with tree-structured syntax (single-bit alphabet).
- 4 intra prediction modes for 16×16, 10 for 4×4 subblocks.
- Simple loop filter with per-edge strength adaptation.
- Single reference frame for WebRTC (plus a "golden" frame for screen content).

VP8's compression efficiency is roughly equivalent to H.264 Constrained Baseline. It served as the baseline WebRTC codec—supported by Chrome and Firefox since WebRTC 1.0—but has been superseded by VP9 in all modern deployments.

B.9.2 VP9 (2013)

VP9 introduced a superblock architecture and several tools that brought it within range of HEVC:

- 64×64 superblocks with recursive quad-tree partitioning down to 4×4.
- 10 intra prediction modes.
- Transforms: DCT and ADST applied independently per row and column, at sizes from 4×4 to 32×32.
- Switchable interpolation filters: 8-tap Regular, Smooth, and Sharp, selected per block.
- Compound prediction: weighted average from two reference frames.
- Segmentation: per-segment QP adjustment for region-of-interest coding.

VP9 achieves approximately 35% rate reduction over H.264 (eq. **??**), placing it at $\Delta_{\mathrm{VP9}} \approx 0.8$. VP9 is the dominant WebRTC codec in Chrome and Firefox as of March 2026, and its temporal SVC support (Section **??**) is the primary mechanism enabling adaptive quality in multi-participant calls.

B.9.3 AV1 (2018)

AV1 represents the most ambitious single-generation improvement in the royalty-free family. Its tool set approaches VVC in breadth while maintaining royalty-free status:

- 128×128 superblocks with flexible partitioning (quad-tree plus rectangular splits) down to 4×4.
- 56+ directional intra prediction modes plus non-directional modes (Paeth, Smooth variants, DC).
- Full inter prediction toolkit: OBMC, warped motion (affine), compound modes (Wedge, Difference-weighted), inter-intra compound prediction.
- 16 transform type combinations (4 1D types $\times$ rows/columns), at sizes up to 64×64.
- Film Grain Synthesis: the encoder characterizes noise/grain parameters and transmits them as metadata; the decoder re-synthesizes grain after reconstruction. This removes non-compressible high-frequency noise from the coding loop—a major bitrate saving for camera-captured content.

- Multi-stage in-loop filtering: deblocking → CDEF → loop restoration (Wiener or Self-Guided), as described in Section **??**.
- Multi-symbol arithmetic coding with CDF-based context adaptation (Section **??**).

AV1 achieves approximately 30% rate reduction over VP9 (eq. **??**), a cumulative ∼54% over H.264, placing it at $\Delta_{\text{AV1}} \approx 0.5$ with current hardware encoders and approaching 0.4 by 2030 as encoder implementations mature.

AV1's tools directly explain why its Δ is smaller than VP9's: more transform types capture residual structure better (reducing $\Delta_{\text{transform}}$), warped and compound motion reduce prediction residual energy (reducing Δ_{pred}), and film grain synthesis removes non-compressible variance from the coding loop (effectively reducing the source variance σ^2 that the codec must represent).

B.10 Block Partitioning: A Generational Comparison

Block partitioning is the most architecturally consequential difference between codec generations. Larger and more flexible partitions allow the encoder to match block boundaries to content structure—smooth regions get large blocks (few bits for partition overhead), detailed regions get small blocks (better prediction accuracy).

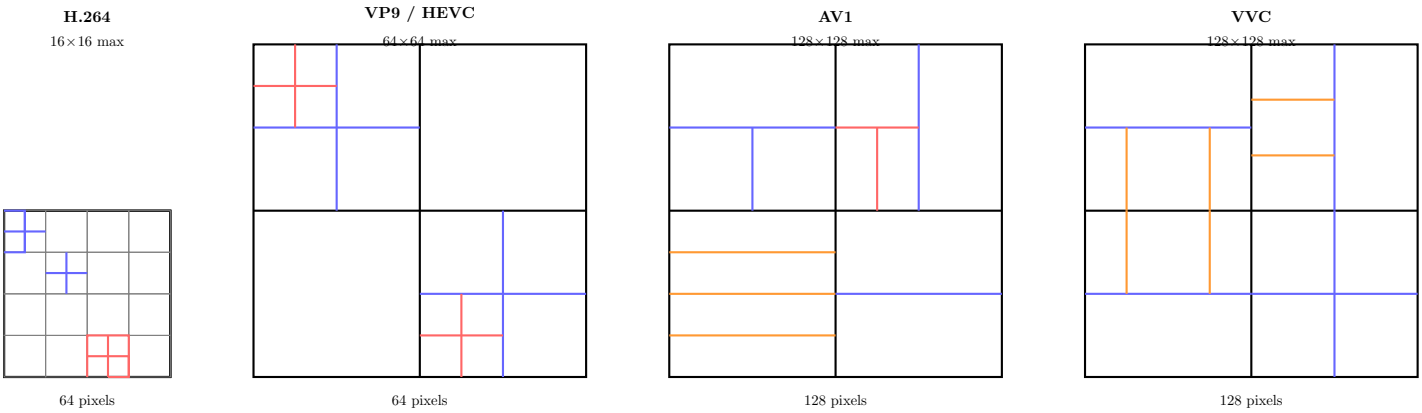


Figure 15: Block partitioning comparison across codec generations. H.264 is limited to 16×16 macroblocks with fixed sub-partitions (left). VP9/HEVC use quad-tree partitioning within 64×64 superblocks (center-left). AV1 extends to 128×128 with quad-tree plus rectangular splits (center-right). VVC adds binary and ternary splits for maximum flexibility (right). Blue lines show first-level splits; red shows deeper partitioning; orange shows asymmetric/ternary splits.

B.11 Rate-Distortion Operating Points by Codec

The codec-specific analyses of Sections **??**–**??** can be unified through the Bjøntegaard Delta Rate (BD-Rate), the standard metric for comparing codec efficiency. BD-Rate computes the average percentage rate difference between two codecs at equivalent distortion by integrating the difference between their rate-distortion curves fit as cubic polynomials in PSNR-vs-log-rate space:

$$\text{BD-Rate} = \frac{1}{\Delta\text{PSNR}} \int_{\text{PSNR}_{\text{min}}}^{\text{PSNR}_{\text{max}}} [\ln R_A(p) - \ln R_B(p)]\, dp \tag{57}$$

A negative BD-Rate indicates that codec A requires fewer bits than codec B at the same quality.

Feature	H.264	VP9	HEVC	AV1	VVC
Max block	16×16	64×64	64×64	128×128	128×128
Min block	4×4	4×4	8×8	4×4	4×4
Split types	Fixed partitions	Quad-tree	Quad-tree	Quad + rect	Quad + BT + TT
Asymmetric	Limited	No	No	4:1 rectangles	Full (BT/TT)

Table 6: Block partitioning capabilities by codec. BT = binary tree (horizontal or vertical), TT = ternary tree (1:2:1 split). Larger maximum blocks reduce overhead for homogeneous regions; asymmetric splits improve partitioning efficiency at content boundaries.

Codec	BD-Rate vs. H.264	Δ (720p)	Family	Year
VP8	∼0%	∼2.1	Google	2010
VP9	−35%	∼0.8	AOM	2013
HEVC	−40%	∼0.7	MPEG/ITU	2013
AV1	−54%	∼0.5	AOM	2018
VVC	−63%	∼0.3	MPEG/ITU	2020

Table 7: BD-Rate savings and coding efficiency gap Δ (eq. **??**) for each codec relative to H.264 at 720p/30fps. The two families (MPEG/ITU and Google/AOM) achieve comparable efficiency within each generation; AV1 slightly trails HEVC, VVC leads AV1 by ∼20%.

Table **??** provides the DSP-grounded explanation for the codec ordering in eq. (**??**) and the savings figures in eqs. (**??**)–(**??**): each generation adds tools that reduce Δ_{pred}, $\Delta_{\text{transform}}$, and Δ_{quant} in (**??**), with diminishing returns as the Shannon bound is approached.

B.12 Scalable Video Coding and Simulcast

Multi-participant WebRTC sessions require adaptive quality: different receivers have different bandwidths, and a single encode rate cannot serve all. Two architectures address this.

B.12.1 Simulcast

Simulcast encodes the source at N independent resolution/bitrate combinations:

$$R_{\text{simulcast}} = \sum_{i=1}^{N} R_i \tag{58}$$

Each encoding is a fully independent bitstream. A Selective Forwarding Unit (SFU) selects which stream to forward to each receiver based on reported bandwidth. The overhead is substantial: a typical 3-layer simulcast (720p/1.5 Mbps, 360p/500 kbps, 180p/150 kbps) requires 2.15 Mbps total—approximately 1.4× the cost of the highest layer alone—with no inter-layer compression benefit.

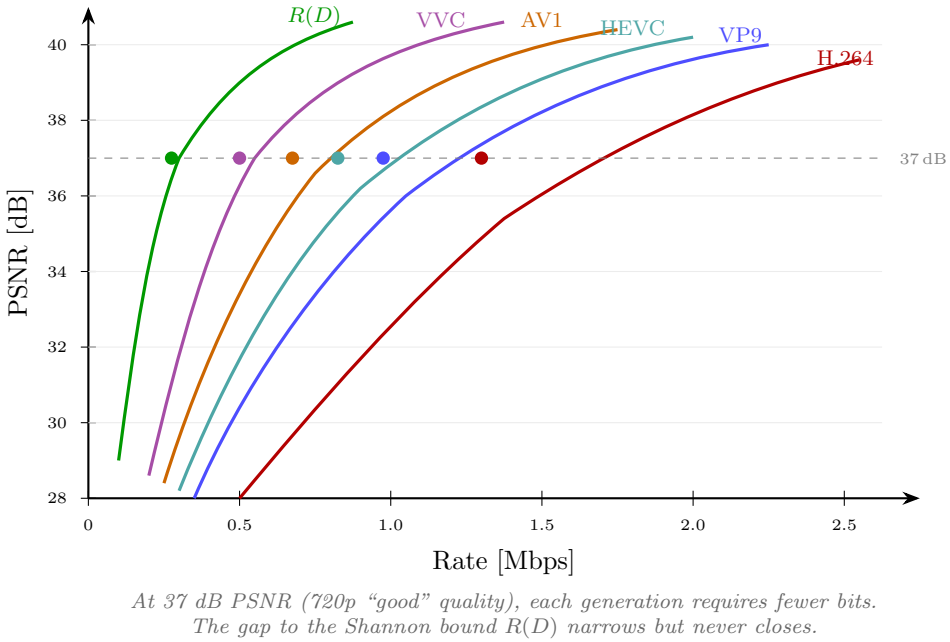


Figure 16: Rate-distortion curves for six codecs at 720p/30fps, with the Shannon $R(D)$ bound (green). At constant quality (37 dB dashed line), the rate progression from H.264 through VVC illustrates the BD-Rate savings in Table **??**. This figure extends Figure **??** from Appendix **??** with the full codec lineage.

B.12.2 Scalable Video Coding (SVC)

SVC produces a layered bitstream where a base layer L_0 is independently decodable and enhancement layers $L_1, \ldots, L_K$ refine quality progressively. Three scalability dimensions exist:

Temporal scalability. The base layer encodes at a reduced frame rate (e.g., 7.5 fps); enhancement layers add intermediate frames:

$$R_{\text{temporal}} = R_0 + \sum_{k=1}^{K} \Delta R_k \tag{59}$$

where $\Delta R_k \ll R_0$ because enhancement frames reference lower layers and carry primarily motion-compensated residuals. An SFU drops enhancement layers for bandwidth-constrained receivers, gracefully degrading frame rate while preserving spatial quality.

Spatial scalability. The base layer encodes at reduced resolution; enhancement layers add detail via inter-layer prediction:

$$R_{\text{spatial}} = R_0 + \Delta R_{\text{upscale}} \tag{60}$$

where $\Delta R_{\text{upscale}}$ exploits correlation between the upsampled base layer and the full-resolution source.

Quality (SNR) scalability. Refinement of quantization at the same resolution, encoded as the difference between coarse and fine reconstructions.

SVC achieves 20–30% rate savings over simulcast for equivalent adaptivity because enhancement

layers exploit inter-layer prediction:

$$R_{\text{SVC}} < R_{\text{simulcast}} \quad \text{(typically by 20–30\%)} \tag{61}$$

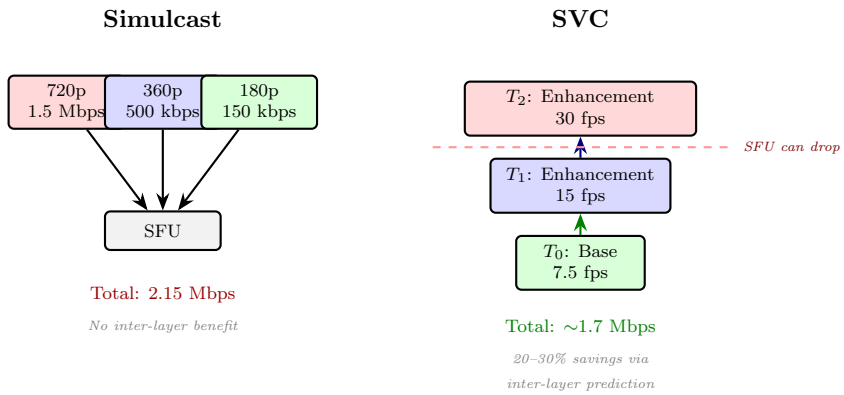


Figure 17: Simulcast (left) encodes three independent bitstreams with no inter-layer compression. SVC (right) encodes a layered bitstream where temporal enhancement layers (T_1, T_2) reference the base layer (T_0), achieving 20–30% savings. The SFU drops enhancement layers for bandwidth-constrained receivers.

Codec-specific SVC support relevant to WebRTC:

- **H.264 SVC (Annex G).** Defines temporal, spatial, and quality scalability. Complex to implement; rarely used in WebRTC deployments.
- **VP9 SVC.** Temporal scalability is widely deployed in WebRTC via libvpx (up to 3 temporal layers). Spatial SVC is defined but implementation-limited. VP9 temporal SVC is the primary mechanism enabling adaptive quality in Chrome/Firefox SFU-based calls.
- **AV1 SVC.** Temporal scalability supported in both libaom and SVT-AV1. A key enabler for AV1 adoption in production WebRTC infrastructure, as SFUs require layered bitstreams to serve heterogeneous receivers.

Safari's lack of simulcast and SVC support in 2021 (Section 3.1) meant that Safari users in multi-participant calls could not benefit from adaptive quality—every receiver got the same stream regardless of bandwidth, degrading the experience for all participants.

B.13 Hardware vs. Software Encoding

The main paper identifies hardware encoding as the deployment gate for AV1 (Sections 8.5, 9.5). This section quantifies why.

B.13.1 Encoding Complexity

The computational cost of encoding a video frame is dominated by motion estimation and RD-optimized mode decision. We express complexity as operations per pixel:

$$C_{\text{enc}} = C_{\text{ME}} + C_{\text{transform}} + C_{\text{quant}} + C_{\text{entropy}} + C_{\text{mode}} \tag{62}$$

Representative values for real-time presets:

- H.264 Constrained Baseline ("veryfast"): $C_{\text{enc}} \approx 200$ ops/pixel
- VP9 (speed 8, real-time): $C_{\text{enc}} \approx 600$ ops/pixel
- AV1 (speed 10, real-time): $C_{\text{enc}} \approx 2000$ ops/pixel

The real-time constraint requires:

$$C_{\text{enc}} \times W \times H \times \text{fps} \leq F_{\text{available}} \tag{63}$$

where $F_{\text{available}}$ is the available computational throughput (operations per second). For 720p/30fps AV1 at 2000 ops/pixel:

$$2000 \times 921{,}600 \times 30 = 55.3 \times 10^9 \text{ ops/s} \tag{64}$$

This consumes approximately one full core of a modern desktop CPU. On mobile devices with thermal and power constraints, software AV1 encoding at 720p/30fps remains impractical without hardware acceleration.

B.13.2 Hardware Encoder Architecture

Hardware encoders implement a fixed subset of the codec's tool set in dedicated silicon. The trade-off is quality for throughput: hardware encoders typically produce output 10–20% larger than software encoders at equivalent visual quality because they implement fewer mode decision candidates and simpler motion search:

$$\Delta_{\text{HW}} \approx (1.1\text{–}1.2) \cdot \Delta_{\text{SW}} \tag{65}$$

This quality penalty is acceptable for real-time communication because: (a) the alternative—software encoding at a lower speed preset or reduced resolution—incurs a larger quality penalty, and (b) the freed CPU capacity can be used for pre/post-processing, noise reduction, or additional application logic.

B.13.3 Hardware Encoder Availability

Platform	**H.264 HW**	**VP9 HW encode**	**AV1 HW encode**	**AV1 year**
Intel (QSV)	Sandy Bridge+ (2011)	Kaby Lake+ (2017)	Arc / Meteor Lake+	2022
NVIDIA (NVENC)	Kepler+ (2012)	—	Ada Lovelace+	2022
AMD (VCN)	VCN 1.0+ (2017)	—	VCN 4.0+ (RDNA 3)	2023
Apple (VideoToolbox)	A4+ (2010)	—	M3+	2023
Qualcomm	SD 820+ (2016)	—	SD 8 Gen 2+	2023

Table 8: Hardware encoder availability timeline. H.264 hardware encoding has been ubiquitous for over a decade. AV1 hardware encoding began shipping in 2022–2023 across all major platforms; the main paper projects it becomes the practical WebRTC default by 2028–2030 as these chipsets reach market saturation.

Excursus: WebCodecs and hardware acceleration. — When a WebCodecs `VideoEncoder` is configured with `codec: "av01"`, the browser selects hardware or software encoding based on platform capability and the `hardwareAcceleration` preference (`"prefer-hardware"`, `"prefer-software"`, or `"no-preference"`). The application receives encoded `EncodedVideoChunk` objects either way, but latency and CPU cost differ dramatically: hardware encoding typically adds < 1 ms per frame and consumes near-zero CPU, while software AV1 at real-time presets consumes an entire CPU core (eq. **??**). This abstraction is what makes the hardware timeline in Table **??** directly relevant to the WebCodecs adoption curve in Section 7.

The hardware encoding timeline directly underpins the main paper's projections: the "AV1 becomes practical" milestone (Section 8.5) and "AV1 HW ubiquitous" milestone (Section 9.5) depend on the chipset generations in Table **??** reaching market saturation.

B.14 Implications for Real-Time Web Communication

The DSP analysis of this appendix maps back to concrete implications for the technologies surveyed in the main text.

The hybrid coding framework is universal; innovation is incremental. Every codec from VP8 to VVC implements the same prediction–transform–quantization–entropy framework (Figure **??**). Generational improvements come from expanding the tool set within this framework—more block sizes (Table **??**), more prediction modes, more transform types (Table **??**)—not from fundamentally new architectures. This means the efficiency gap Δ approaches but never reaches zero: each additional tool yields diminishing rate-distortion improvement at increasing computational cost.

Transform and prediction innovations explain the BD-Rate progression. The 35% rate saving from H.264 to VP9 comes primarily from larger blocks (64×64 vs. 16×16) and adaptive transforms (DCT+ADST vs. fixed DCT). The additional 30% from VP9 to AV1 comes from compound and warped motion, film grain synthesis, and 16-way transform type selection (Table **??**). Each tool adds encoding complexity proportional to its rate-distortion benefit, which is why real-time encoders must aggressively prune the tool set.

SVC is what makes multi-participant WebRTC work. Without temporal SVC, an SFU must forward full-rate streams to all participants regardless of their bandwidth. With VP9/AV1 temporal SVC (Section **??**), the SFU drops enhancement temporal layers for bandwidth-constrained receivers, degrading frame rate gracefully while preserving spatial quality. The 20–30% savings over simulcast (eq. **??**) translate directly to reduced server egress bandwidth in large meetings.

Hardware encoding is the deployment gate, not codec specification. AV1 was specified in 2018 but becomes a practical WebRTC default only when hardware encoders are ubiquitous (∼2028–2030). The DSP tools that make AV1 efficient—warped motion, OBMC, 128×128 superblocks with flexible partitioning—are precisely the tools that make it computationally expensive in software (eq. **??**). The ∼10–20% quality penalty of hardware encoders (eq. **??**) is the price of real-time feasibility, and it is a price worth paying: $\Delta_{\text{AV1,HW}} \approx 0.6$ is still far below $\Delta_{\text{VP9,SW}} \approx 0.8$.

The convergence toward Shannon is real but asymptotic. Cross-referencing with Appendix **??**: H.264's $\Delta \approx 2.1$ means it operates at 3.1× the theoretical minimum rate. VP9's $\Delta \approx 0.8$ means 1.8×. AV1's $\Delta \approx 0.5$ means 1.5×. VVC's $\Delta \approx 0.3$ means 1.3×. Each generation

captures more spatial, temporal, and perceptual redundancy, but the returns are diminishing: the distance from H.264 to VP9 (-35%) exceeds VP9 to AV1 (-30%), which exceeds AV1 to VVC (-17%). The next generation of efficiency gains for real-time web communication will come not from codecs alone, but from the co-optimization of transport (QUIC, WebTransport), application architecture (WebCodecs, unbundled pipelines), and network infrastructure that the main text describes.